Why Temporal Abstraction Can Fail in Imagination-Based RL: A Diagnostic Analysis of Variable-Rate DreamerV3

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Abstract: World-model reinforcement learning is increasingly deployed for robot control, where temporal abstraction enables planning across heterogeneous time scales. We find that Δt-conditioning, effective in model-predictive control (MPC), cannot transfer to imagination-based actor-critic training: every variable-rate DreamerV3 configuration degrades to 14–19% of baseline. Factorial diagnostics on two continuous-control tasks isolate trained temporal embedding with multi-rate data exposure as the strongest predictor of this failure—*conditioning-induced value overestimation* (CIVO; 26/29, AUROC$=0.953$, LOEO cross-validated). The RSSM develops a $2.78\times$ reward prediction gap at $\Delta t{>}1$ that the critic internalizes as optimistic values; the contrast with MPC identifies feedback architecture as the structural factor. No tested intervention across four levels (gradient, output, representation, architecture) reliably eliminates CIVO: dual critics converge in pure imagination ($\rho > 0.999$), rendering Double Q pessimism ineffective, though min + increased capacity enables recovery. A lightweight NVE detector achieves 100% sensitivity.

Keywords: World Models, Temporal Abstraction, Value Overestimation, DreamerV3

1 Introduction

World-model reinforcement learning enables sample-efficient learning directly on physical hardware [1, 2]. Real robotic tasks demand temporal abstraction: manipulation requires fast reactive grasping alongside slow approach planning, locomotion involves gait phases at different frequencies, and sim-to-real transfer must handle dt mismatches between simulation and hardware. TAWM [3] shows that conditioning a world model on step size dt yields consistent gains within TD-MPC2's MPC framework [2]. Can the same dt-conditioning bring temporal abstraction to imagination-based actor-critic methods like DreamerV3 [4], where policies are trained entirely in imagined rollouts?

We find that it cannot. Every variable-rate configuration degrades performance catastrophically (14–19% of baseline). Through a factorial analysis on Walker Walk [5], we isolate the combination of a *trained* dt-embedding and multi-rate data exposure as the strongest predictor. This combination enables the RSSM reward head to develop a $2.78\times$ prediction gap between $\mathrm{dt}{=}1$ and $\mathrm{dt}{=}2$, which the critic internalizes as optimistic value estimates, a failure mode we term *conditioning-induced value overestimation* (CIVO), analogous to delusional bias [6] but model-induced rather than policy-class-induced. A normalized value-to-eval (NVE) threshold of $\geq 1.5\times$ achieves 26/29 classification ($n{=}29$, two environments, AUROC$=0.953$, in-sample; LOEO cross-validated, Appendix Table 10).

Gradient decomposition reveals that gradient magnitude does not predict causal importance: the reward head contributes 38–49% of $\nabla_\theta e_{\mathrm{dt}}$ yet reward-only stopping fails, while the observation decoder (16–22%) carries the critical signal. A four-level intervention hierarchy (gradient, output, rep-

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resentation, architecture) shows that no tested intervention reliably eliminates CIVO across environments. In pure-imagination training, dual critics converge to identical value functions ($\rho > 0.999$), rendering Double Q pessimism ineffective; the min operation nonetheless enables CIVO recovery, while prevention requires min + increased capacity. These findings are directly relevant to robotics: as world-model RL moves toward real systems with variable-rate sensing and multi-timescale control, understanding when temporal conditioning fails, and why standard countermeasures cannot fix it, is essential for safe deployment.

Contributions.

1. **Failure mode identification.** We identify CIVO as specific to imagination-based actor-critic with temporal conditioning and isolate trained dt-embedding + multi-rate exposure as jointly sufficient in tested configurations (6/6 seeds, two locomotion environments, AUROC$=0.953$, in-sample; LOEO cross-validated). Cross-environment validation spanning locomotion and manipulation (ManiSkill) establishes that CIVO is locomotion-specific; frozen-embedding controls on three manipulation tasks identify value-ratio transients as environment-driven, demonstrating the diagnostic's specificity on non-CIVO dynamics.
2. **Architectural boundary.** dt-conditioning succeeds in TD-MPC2's MPC re-planning but fails under imagination-based bootstrapping, where model errors compound into biased value targets with no real-data anchor.
3. **Four-level intervention hierarchy.** Across gradient, output, representation, and architecture levels, no intervention reliably eliminates CIVO across environments. Gradient magnitude does not predict causal importance (reward head: 38–49%, reward-only stopping fails). In pure-imagination training, dual critics converge ($\rho > 0.999$), eliminating Double Q pessimism, though the min operation enables recovery through a different mechanism.
4. **Online detection.** A lightweight NVE-based detector achieves 100% sensitivity (5/5) with predictive warning in 2/5 cases.

2 Related Work

Temporal abstraction in RL. The options framework [7], option-critic [8], HAM [9], and MAXQ [10] formalize temporal abstraction through hierarchical policies. Semi-MDP theory [11] provides the mathematical foundation. Action-repetition methods [12, 13, 14] achieve simpler temporal abstraction via real transitions; Wang and Beltrame [15] learn elastic time steps that adapt step size to state dynamics. The Dreamer family [16, 17, 4, 18] trains policies in imagination at fixed $\mathrm{dt}{=}1$; hierarchical variants [19, 20, 21, 22] use subgoals or separate dynamics per scale, avoiding flat dt-conditioning. TAWM [3] conditions a world model on dt within TD-MPC2 [2] and achieves consistent gains; the critical distinction is that MPC re-plans from real states, so model errors do not accumulate into a critic (§5).

Value overestimation and model fidelity. Maximization bias [23] has been addressed by Double Q-learning [24], TD3 [25], and ensemble methods [26]; in model-based settings, MBPO [27] and MOPO [28] bound rollout error, while MOReL [29] and COMBO [30] penalize model uncertainty. Lu et al. [6] identify delusional bias from policy-class constraints. CIVO is mechanistically distinct: the amplifier is the world model's co-adaptation with a learned temporal embedding, producing systematic reward-prediction bias at $\mathrm{dt} > 1$. Multi-timestep world models [31] learn predictions at variable horizons but do not examine the value overestimation pathology we isolate. Compounding error in multi-step rollouts [32, 33] is a related but distinct phenomenon: if CIVO were generic compounding error, untrained dt-embeddings at $\mathrm{dt}{=}2$ should also produce overestimation, yet they do not (NVE$<0.6\times$ for 008a/008c vs. $4.12\times$ for 008d). Furthermore, signed reward error analysis shows $\mathrm{dt}{=}2$ rewards are *under*predicted (mean SE$=-0.599$), ruling out observation-level overprediction and localizing CIVO to imagination-level prior drift (§5). Compounding error decision-aware model learning [34] optimizes for planning performance but does not address dt-conditioned co-adaptation. DreamSmooth [35] smooths reward prediction targets but does not

address the fidelity gap at $\mathrm{dt} > 1$. Our TCN intervention relates to training-with-noise [36] and NoisyNet [37]; PETS [38] and STEVE [39] use ensemble disagreement for epistemic uncertainty. TCN's environment-specific dose sensitivity (§4.5) suggests CIVO is a structural vulnerability, not a regularization deficiency. Our factorial diagnostic follows Henderson et al. [40].

3 Experimental Setup

3.1 Base Architecture

All experiments build on DreamerV3 [4], which learns a recurrent state-space model (RSSM) with discrete latent states (32 categories $\times$ 32 classes), uses symlog-transformed predictions, and trains an actor-critic entirely within imagined trajectories. The standard configuration imagines $H{=}15$ steps at a fixed temporal resolution of $\mathrm{dt}{=}1$, meaning each imagination step corresponds to one environment transition.

3.2 Variable-Rate Imagination Mechanism

We introduce three orthogonal architectural modifications that enable variable-rate imagination in DreamerV3. Each can be independently toggled to isolate its contribution.

Temporal embedding. A learned embedding $e(\mathrm{dt})$ is concatenated to the RSSM input at each imagination step to condition state predictions on the temporal scale. We test three modes:

- **Trained**: embedding parameters are updated via backpropagation through the world model loss, so the RSSM can co-adapt its dynamics and reward predictions to the temporal scale.
- **Frozen**: a fixed random projection of identical dimensionality (`requires_grad=False`). The RSSM receives a temporal signal but cannot co-adapt to it.
- **Removed**: embedding dimension set to zero; no temporal conditioning signal is provided.

We further distinguish **trained-and-exposed** (subsample ON: the RSSM trains on $\mathrm{dt}{>}1$ transitions) from **trained-but-unexposed** (subsample OFF: $\mathrm{dt}{>}1$ used only during imagination). This four-way classification (trained/exposed, trained/unexposed, frozen, removed) is the key experimental structure of our factorial design.

Temporal subsampling. When enabled, the replay buffer augments training data with variable-dt transitions by subsampling consecutive frames. Concretely, with a maximum subsampling factor of 2, the RSSM trains on both $\mathrm{dt}{=}1$ and $\mathrm{dt}{=}2$ transitions drawn from the same episodes. When disabled, only standard $\mathrm{dt}{=}1$ transitions are used. This factor controls whether the world model has seen multi-scale dynamics during training.

Imagination horizon and step size. The imagination horizon H sets the number of rollout steps. We test $H{=}15$ (default) and $H{=}8$. The reduced horizon controls for compounding error: $H{=}8$ steps at $\mathrm{dt}{=}2$ covers 16 physical time steps, comparable to the 15 physical steps of the default configuration. The parameter `gate_fixed_dt` forces all imagination steps to use a specific dt value (1, 2, or 4); this separates the effect of dt-conditioning from any learned gating policy.

3.3 Factorial Design

We adopt a factorial ablation across four factors: dt-embedding mode (trained / frozen / none), temporal subsampling (on / off), imagination horizon ($H{=}15$ / $H{=}8$), and imagination step size ($\mathrm{dt}{=}1$, 2, 4). The design (12 conditions, Table 12 in Appendix C) isolates trained dt-embedding and multi-rate data exposure as orthogonal factors. Key configurations use 4 seeds; following Agarwal et al. [41], we report per-seed trajectories.

3.4 Diagnostic Metrics

Normalized value-to-eval ratio (nve). We define $\text{nve}_t = (v/e)_{\text{config},t}/(v/e)_{\text{baseline},t}$, where v/e is the critic's mean imagined value divided by mean evaluation return, and the baseline is the highest-performing seed (exp_001, seed 1, 505K steps). We restrict to a **35K+ window** to avoid early-training inflation. The $\text{nve} \geq 1.5\times$ threshold achieves 26/29 accuracy (AUROC=0.953, in-sample; LOEO cross-validated, mean AUROC=0.981, Appendix Table 10).

3.5 Selective Gradient Stopping

To identify which gradient pathway through $e_{\Delta t}$ drives CIVO, we introduce *selective gradient stopping* (SGS): $e_{\Delta t}$ is detached via $\text{sg}(\cdot)$ before selected prediction heads during the backward pass, preventing those loss gradients from updating $e_{\Delta t}$ while preserving its forward-pass contribution. We test three variants: reward-only (exp_016e), full-pathway (exp_018, all non-KL gradients blocked), and obs-decoder-only (exp_025/026). Gradient decomposition using $\nabla_\theta \mathcal{L}_i$ quantifies each pathway's contribution (Appendix D.2).

3.6 Forward-Pass Interventions

We test two forward-pass interventions as representation- and output-level probes: **TCN** adds Gaussian noise $\mathcal{N}(0, \sigma^2)$ to $e_{\Delta t}$ during imagination ($\sigma \in \{0.5, 0.2\}$, three environments); **RAVE** clips imagined values exceeding an EMA anchor ($c \in \{2.0, 1.5\}$).

3.7 Dual-Critic Mechanism Ablation

To test whether standard overestimation countermeasures address CIVO, we conduct a 2×2 factorial: {single, dual} $\times$ {narrow ($\sim$1.2M), wide ($\sim$2.4M)} critic parameters. The dual-critic variant follows TD3 [25]: two independent value heads trained on shared RSSM features, with the actor optimizing $\min(v_1, v_2)$. The wide variant doubles hidden dimensions to control for capacity independently of the min operation.

3.8 Environment and Training Details

All experiments use Walker Walk and Cheetah Run from the DeepMind Control Suite (DMC; Tassa et al. 5) with proprioceptive observations, chosen for their contrasting dynamics (6-joint biped vs. 2-joint runner). Walker Walk serves as the primary environment with the full factorial design; Cheetah Run serves as a severity-dependent generalization probe: CIVO manifests at 8–70$\times$ NVE on Cheetah versus 2–4$\times$ on Walker, testing whether interventions transfer under amplified severity rather than replicating the full factorial. The baseline runs for 500K steps (peak return 970.4); treatments run for 105K steps, sufficient to observe collapse onset. All hyperparameters follow the DreamerV3 defaults [4] without task-specific tuning (d=32 temporal embedding; full hyperparameters in Appendix E). Training uses a single NVIDIA RTX PRO 6000 GPU ($\sim$80 GPU-hours total). The codebase extends the official DreamerV3 implementation[1] with the variable-rate modules described in §3.2.

4 Results

We organize results around seven findings (F1–F7) that establish a causal chain from trained temporal conditioning to pathological value dynamics. All experiments use Walker Walk (DMC) with DreamerV3; treatments run to 105K steps and the baseline to 505K steps. We report evaluation returns (mean over 10 episodes) and the normalized value-to-eval ratio (nve, §3.4).

[1] https://github.com/danijar/dreamerv3, accessed March 2026.

Table 1: Key configurations and outcomes. NVE reports peak nve in the 35K+ window (§3.4). CIVO-positive seeds are shaded. Complete results for all 43 configurations in Table 3.

Exp	Config	dt_emb	Sub	H	Eval@final	NVE	Outcome
001	Baseline (dt=1)	—	—	15	915.7 @105K	1.00×	Healthy
008d	Factorial (s1)	trn	ON	8	82.8 @55K	4.12×	Full collapse
008f	No dt_emb (s1)	—	ON	8	503.0 @105K	0.97×	Healthy
008a	Unexposed emb	exp[a]	OFF	15	777.7 @105K	0.93×	Healthy
015b	Frozen emb	frz[b]	ON	8	204.2 @105K	1.45×	Grad. starvation
022b	Frozen emb + Gumbel gate	frz[b]	ON	8	300.7 @105K	0.63×	No CIVO
Cross-environment validation							
010b	Cheetah Run	trn	ON	8	51.7 @105K	8.27×	Collapsed
019b	PickCube-v1	trn	ON	8	26.64 @200K	3.78×[c]	Transient onset
049	StackCube-v1	trn	ON	8	34.9 @212K	5.05×[d]	Env. transient
050	PegInsertionSide-v1	trn	ON	8	20.86 @212K	10.05×[e]	Env. dynamics[e]
046	Cheetah Run	frz[b]	ON	8	150.5 @215K	1.35×	No CIVO

trn = trained (d=32, subsample ON); exp[a] = trained, subsample OFF; frz[b] = frozen (d=32, `requires_grad=False`); — = removed (d=0).
[c] NVE normalized against exp_019a_v2 (vanilla DreamerV3 on PickCube-v1).
[d] StackCube transient: frozen control (exp_051) exhibits identical spike (peak 4.30× at 42.5K, self-correcting by 82.5K), indicating environment-specific phenomenon unrelated to dt-embedding training.
[e] Seed 0 trained peaks at 10.05× (82.5K) with delayed self-correction; seed 1 trained (exp_055) peaks at 3.08× (92.5K), final 0.50×; seed 1 frozen (exp_056) peaks at 11.32× (82.5K), final 1.19×. Environment-specific stochastic dynamics: frozen final exceeds trained final.

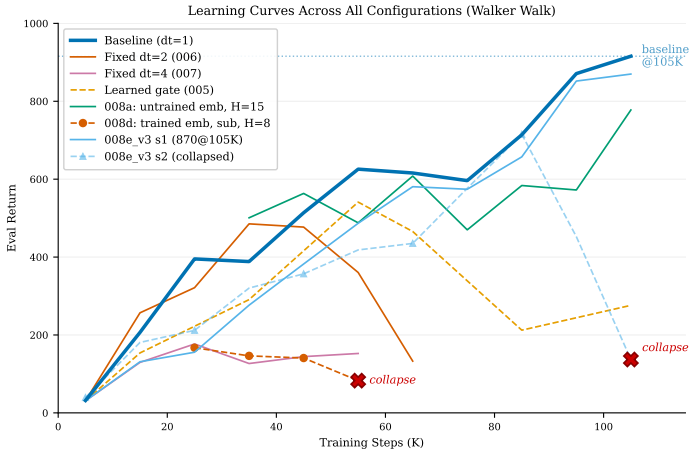


Figure 1: Learning curves. The baseline (black) reaches 915.7; all variable-rate configurations degrade.

4.1 F1: All Variable-Rate Configurations Degrade

Every configuration that introduces $\mathrm{dt} > 1$ imagination degrades relative to the dt=1 baseline (Figure 1). The baseline reaches 915.7 at 105K; fixed dt=2 (006) peaks at 485.3 before collapsing to 132.9 (14% of baseline); fixed dt=4 (007) never exceeds 176.9 (19%); the learned gate (005) collapses to 98.8 (11%).

4.2 F2: Trained Temporal Embedding Precedes Value Overestimation

The factorial design isolates trained dt-embedding as the strongest predictor. The *primary* comparison—008d (trained, subsample ON) vs. 008f (no embedding, subsample ON)—differs only in embedding training, yielding $\mathrm{nve} = 4.12\times$ (eval 82.8) vs. $\mathrm{nve} \leq 0.97\times$ (eval 503.0). *Secondary* comparisons against 008a/008c (unexposed embedding) co-vary subsample status and hori-

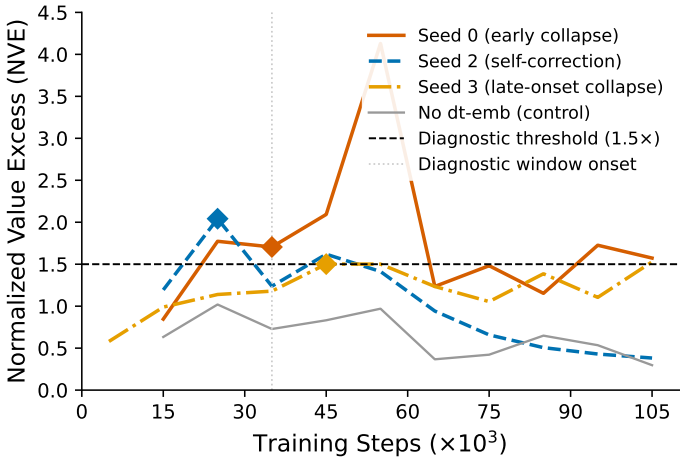


Figure 2: NVE trajectories (mean ± seed range). Trained dt-embedding (red, Walker Walk, n=3 seeds plotted; seed 4 data in Table 3) exceeds the 1.5× threshold (dashed) in all seeds; no-embedding control (blue) remains near 1.0×. Three trajectory types emerge: early collapse (seed 0), late-onset collapse (seed 3), and self-correction (seed 2). Dotted line marks the 35K diagnostic window onset.

zon alongside the embedding factor. The nve $\geq 1.5\times$ threshold (35K+ window) correctly classifies 26 of 29 experimental configurations across two environments (AUROC=0.953, in-sample; LOEO cross-validated, Table 10; Figure 2).

The trained dt-embedding precedes CIVO in all four Walker Walk seeds (peak NVE: 4.62×, 4.12×, 1.62×, 1.53×), universally degrading performance to below 30% of baseline (IQM 175.2, mean 170.6, n=4; Agarwal et al. 41). Downstream trajectories diverge: two seeds undergo full collapse, one exhibits late-onset collapse still declining at 105K, and one self-corrects to a stable but degraded state (29.3% of baseline). Combined with Cheetah Run (§4.4), CIVO onset is universal across all six tested seeds (n=6, two environments) with 5/6 progressing toward collapse (Appendix D.3).

4.3 F3–F5: Factorial Ablation (Appendix C)

Trained dt-embedding and multi-rate data exposure are *jointly sufficient in tested configurations* for CIVO. Neither factor alone precedes pathological overestimation: trained embedding without multi-rate exposure (008a, 3 seeds, peak NVE 0.83–0.93×) and multi-rate exposure without embedding (008f, 3 seeds, peak NVE $\leq 1.41\times$) both remain below 1.5×. Cross-environment replication on Cheetah Run confirms generalization. Freezing the embedding (`requires_grad=False`) dramatically reduces severity (NVE 0.19–1.88× vs. trained 1.53–4.62×; 2/4 seeds below threshold, 2/4 transient self-correcting spikes; Table 11); removing the embedding (008f) also avoids CIVO. Partial gradient blocking (exp_044) does not prevent onset (peak NVE 4.45×) but permits delayed self-correction (final eval 538, 59% baseline; Appendix C), confirming that *any* gradient pathway updating e_{dt} suffices to induce CIVO. Degradation scales monotonically with dt (-47% at dt=2, -81% at dt=4; Figure 5).

ManiSkill manipulation tasks. None of three ManiSkill environments exhibit dt-embedding-dependent CIVO. StackCube-v1 value-ratio spikes are environment-specific: trained (049/052, NVE 5.05/4.67×) and frozen (051, 4.30×) match, all self-correcting by 82.5K. PegInsertionSide-v1 exhibits high-amplitude stochastic dynamics (4–11×) whose trained/frozen ordering is seed-dependent: seed 0 trained peaks higher (10.05× vs. frozen 1.82×), but seed 1 reverses (frozen peak 11.32× vs. trained peak 3.08×). PickCube-v1 frozen control (exp_054, peak 8.81× at 62.5K, self-correcting) confirms the same pattern. CIVO is locomotion-specific in DreamerV3; the frozen-embedding diagnostic correctly identifies these transients as non-CIVO.

Table 2: Intervention hierarchy. Prevention = peak NVE $< 1.5\times$ (35K+). No level reliably eliminates CIVO across environments.

Level	Intervention	Seeds	Prev.	Rate
—	Untreated	6	0/6	0%
Gradient	Reward-only SGS	2	0/2	0%
Gradient	Full-pathway SGS	3	0/3	0%
Gradient	Obs-decoder SGS	6	1/6	17%
Output	RAVE (c=1.5,2.0)	2	0/2	0%
Repr.	TCN (σ=0.1)	1	1/1[‡]	100%
Repr.	TCN (σ=0.2)	6[†]	2/6	33%
Repr.	TCN (σ=0.5)	3	0/3	0%
Repr.	TCN (σ=1.0)	1	1/1[§]	100%
Arch.	Wide single	2	0/2	0%
Arch.	Narrow dual (min)	2	0/2	0%
Arch.	Wide dual (min)	4	3/4[‖]	75%
—	Embedding frozen	7	5/7[#]	71%
—	Embedding removed	5	5/5	100%

[†]Walker 2/4 (s2 NVE 2.11×; s3 late-onset CIVO, NVE 3.04× at 105K); Cheetah 0/1, PickCube 0/1. [‡]Borderline (NVE=1.02×, eval 55%). [§]CIVO prevented but progressive policy erosion (peak 116%, final 15% baseline). [‖]Walker 3/3 ($\mathrm{nve} \leq 1.4\times$; 1/3 non-CIVO eval collapse), Cheetah 0/1.
[#]Walker 5 + Cheetah 2. Exp_047 (NVE 1.71×) and Exp_057 (NVE 1.88×): both transient, self-correcting within 20K steps; remaining 5 seeds: NVE 0.19–1.45×.

4.4 F6–F7: Interventions and Multiple Pathways

F6: Trained embedding creates multiple vulnerability pathways. Exp_008e_v3 restricts imagination to $\mathrm{dt}{=}1$ while retaining the trained embedding. Two of four seeds collapse (50%): Pathway A (CIVO, $\mathrm{nve} = 4.02\times$) and preliminary Pathway B (entropy *increase*, $\mathrm{nve} = 3.78\times$; 2 seeds only, requires confirmation; Appendix D.6). Pathway B produces one of three non-CIVO false positives (n=29).

F7: Full-pathway gradient stopping shows mixed results. Full-SGS (exp_018) detaches all non-KL gradients from e_{dt}. Seed 1 shows partial mitigation (eval 448.3, NVE 1.66×); seed 2 collapses (0/2 elimination; Appendix D.2). Cross-environment validation on Cheetah (010b, NVE 8.27–108.52×) confirms CIVO generalization; full-SGS yields 0/3 elimination. Removing the embedding eliminates CIVO on Cheetah (024b, NVE $< 0.2\times$).

Targeted obs-decoder-only SGS confirms pathway redundancy. Obs-decoder-only SGS (exp_025/026) achieves 1/6 CIVO prevention (1/4 Walker, 0/2 Cheetah; Table 2), not meaningfully different from full-pathway SGS (0/3). CIVO operates through redundant gradient channels: blocking any single pathway shifts co-adaptation to the remaining ones (Appendix D.3).

4.5 Forward-Pass Interventions

We test two forward-pass alternatives (Table 2). **RAVE** (value-level capping) fails: the EMA anchor tracks biased values during CIVO onset. **TCN** (representation-level noise) suppresses CIVO in 2 of 4 Walker seeds at σ=0.2 (NVE $< 1.0\times$, eval 45–91%), but remaining seeds show runaway escalation, suggesting TCN gates rather than attenuates CIVO. Cross-environment transfer fails: Cheetah effective dose (σ=1.0) causes policy erosion to 15% baseline; PickCube 0/1 (Appendix D.8). TCN is a *diagnostic probe*, not a deployable mitigation.

4.6 Architecture-Level Ablation

Critic convergence in pure imagination. Across all dual-critic configurations (n=7), critics trained on shared RSSM features converge ($\rho > 0.999$), with pessimism at 3–4% of the independent-critic bound (§3.7).

2×2 factorial: recovery vs. prevention. Min drives recovery: narrow dual-critic ($\sim$1.2M) recovers fully ($\sim$16–19$\times$ recovery ratio) despite $2\times$ fewer parameters. Prevention requires min + capacity: wide dual-critic maintains $\mathrm{nve} \leq 1.4\times$ (3/3 prevention, 1/3 non-CIVO collapse). Cross-environment transfer is limited (Appendix G).

4.7 Online CIVO Detection

We derive a lightweight online detector monitoring the value-to-evaluation ratio for superlinear growth. Applied to 14 runs across three environments, it achieves TPR=$5/5$, FPR=$2/9$, with predictive alerts in 2/5 cases (Appendix F).

5 Discussion

CIVO mechanism. The RSSM develops a $2.78\times$ reward prediction gap under trained conditioning (Table 14), analogous to delusional bias [6]. Gradient magnitude $\neq$ causal importance: the reward head (38–49% of $\nabla_\theta e_{\mathrm{dt}}$) fails when blocked alone; the observation decoder (16–22%) carries the critical signal; full-pathway SGS (0/3) confirms forward-pass conditioning as the irreducible vulnerability. CIVO is upstream of estimation error—dual critics converge (§4.6)—so standard corrections are inapplicable; MPC re-planning prevents accumulation (Appendix D.4).

Practitioner guidelines. Monitor via NVE diagnostic ($\geq 1.5\times$, AUROC=0.953; Table 10). Freezing the dt-embedding converts sustained CIVO (6/6 trained locomotion, NVE 1.53–$108\times$) to transient spikes that self-correct within 20K steps (2/3 frozen Walker, 1.71–$1.88\times$; 0/2 frozen Cheetah); untreated, extreme CIVO ($70.6\times$) causes irrecoverable damage (Appendix D).

Implications for robot learning. No dt-embedding-dependent CIVO was observed in manipulation: all three ManiSkill environments exhibit ve-ratio transients (peaks 4–$11\times$), but frozen controls show comparable or higher peaks (Table 4), establishing environment-specific origin. A possible mediator is reward density: locomotion provides continuous rewards accumulating coherent dt-embedding bias, while manipulation rewards are sparse and contact-gated.

6 Conclusion

We identify trained dt-embedding with multi-rate data exposure as the strongest predictor of CIVO in DreamerV3 (26/29, AUROC=0.953; LOEO cross-validated). No tested intervention reliably eliminates CIVO; in pure imagination, dual critics converge ($\rho > 0.999$), rendering Double Q pessimism ineffective. The feedback architecture—MPC re-planning versus imagination-based bootstrapping—is the structural factor.

Limitations. Our analysis covers proprioceptive locomotion (DMC) and simulated manipulation (ManiSkill) but does not test pixel-based observations, Atari benchmarks, or real-robot deployments where sensor noise and actuation latency introduce additional confounds. All experiments use DreamerV3's RSSM; whether CIVO manifests in other imagination-based architectures (e.g., IRIS [42] or latent-space planners beyond TD-MPC2) remains open. We characterize CIVO under fixed-dt subsampling; learned temporal gates warrant separate study. Statistical power with 4 seeds per configuration is limited [43]; the NVE threshold (LOEO AUROC=0.981) may require recalibration for different reward scales. Code: upon acceptance.

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A Complete Experimental Results

Table 3 reports all 42 experimental configurations. The main text (Table 1) presents the subset most relevant to the causal argument; this table includes the additional variable-rate configurations (006, 007, 005), multi-seed factorial runs (009b, 008d_s3, 008d_s4), SGS interventions (018), cross-environment seeds (010b_s0), and mitigation experiments (008e_v3).

B NVE Threshold Sensitivity Analysis

We report the full sensitivity analysis for the NVE diagnostic framework introduced in Section 4. The main text reports Task B2 (any CIVO detection, $\mathrm{nve} \geq 1.5\times$). Here we additionally present Task B1, which restricts the positive class to *sustained* CIVO, specifically experiments exhibiting persistent, accelerating value overestimation that progresses to performance collapse. Transient CIVO cases (e.g., exp 009b, which self-corrects after a brief excursion above the threshold) are classified as negative in B1.

B.1 Per-Experiment NVE Values

Table 5 lists the NVE values at 35K+ training steps for all ten factorial experiments (including four 008e_v3 seeds). Experiments are sorted by descending NVE. Experiment 008e_v3_s2 exhibits elevated NVE ($3.78\times$) due to Imagination-Reality Divergence (IRD), a mechanistically distinct phenomenon from CIVO, and is classified as negative in both tasks. Experiment 008e_v3_s3 exhibits CIVO with NVE $4.02\times$ despite $\Delta t{=}1$ imagination.

B.2 Multi-Threshold Classification Performance

Table 6 reports Task B1 classification metrics across six NVE thresholds. Task B1 achieves AUROC $= 1.000$ because the minimum positive NVE ($4.02\times$, exp 008e_v3_s3) exceeds the maximum negative NVE ($3.78\times$, exp 008e_v3_s2), yielding perfect rank separation. The optimal threshold lies in the interval $(3.78, 4.02]$; we report $\geq 3.9\times$ as a representative value. Table 7 provides the corresponding analysis for Task B2 (AUROC $= 0.958$, imperfect separation due to 008e_v3_s2).

B.3 Threshold Recommendation

For the main diagnostic (Task B2), we recommend $\mathrm{nve} \geq 1.5\times$ as the operating threshold for three reasons:

1. **Maximum recall.** The $1.5\times$ threshold achieves 100% recall, detecting all four CIVO cases including the transient case (009b, $\mathrm{nve} = 1.62\times$). Raising the threshold to $1.7\times$ loses 009b, dropping recall to 75.0%.
2. **Best F1.** At $1.5\times$, F1 = 88.9%, the highest among all tested thresholds for B2. The single false positive (008e_v3_s2) is mechanistically identifiable as IRD rather than CIVO.
3. **Sensitivity–specificity tradeoff.** Lower thresholds ($1.0\times$) add false positives without gaining true positives. Higher thresholds ($\geq 1.7\times$) sacrifice recall without eliminating the 008e_v3_s2 false positive (which requires $\tau > 3.78$).

For Task B1 (sustained CIVO only), perfect classification is achievable at $\mathrm{nve} \geq 3.9\times$, but with only $n_{\mathrm{pos}} = 3$ the statistical power remains limited (accuracy 95% CI: [69.2%, 100%]).

B.4 Expanded ROC Analysis

The original n=16 analysis (Tables 6–8) predates the SGS, RAVE, and TCN experiments. We extend the NVE diagnostic evaluation to n=29 configurations spanning all tested interventions. Figure 3 shows the receiver operating characteristic curve with 95% bootstrap confidence interval (10,000 resamples).

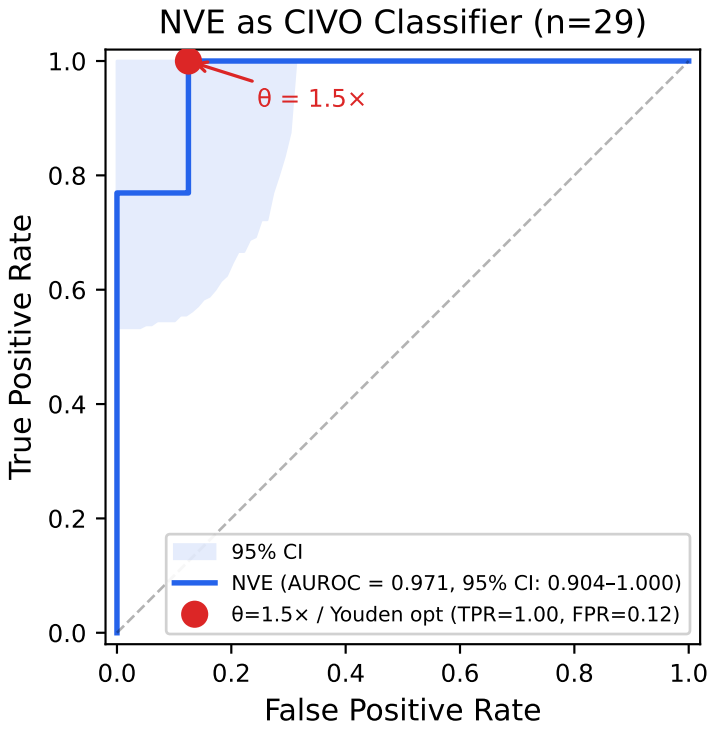


Figure 3: NVE ROC curve (n=29 configurations, AUROC=0.953, in-sample, post-hoc). The preset $\mathrm{nve} \geq 1.5\times$ operating point (star) achieves TPR=1.00, FPR=0.18. Three false positives arise from non-CIVO value overestimation: RAVE anchor pollution, obs-decoder SGS side effects, and IRD eval-driven collapse (008e_v3 seed 2). Threshold calibrated and validated in-sample, post-hoc; cross-architecture validation deferred.

We label a configuration as CIVO-positive if and only if (i) it uses a trained dt-embedding with multi-rate data exposure, and (ii) NVE exceeds $1.5\times$ with co-occurring entropy collapse. Con-

figurations exhibiting high NVE through other mechanisms (gradient truncation side effects, value anchor pollution) are labeled non-CIVO.

The three false positives (FPR$=0.18$) are mechanistically distinct from CIVO: (1) RAVE anchor pollution (exp_028, the EMA anchor becomes inflated, producing overestimation through a different pathway); (2) obs-decoder SGS-induced value distortion (exp_025 seed 2, blocking observation decoder gradients damages the RSSM latent space); (3) IRD eval-driven collapse (exp_008e_v3 seed 2, actor entropy *increases* with evaluation collapse, mechanistically distinct from CIVO's entropy *decrease*). All three produce NVE $> 1.5\times$ through non-CIVO mechanisms, indicating that the NVE diagnostic detects value overestimation broadly rather than CIVO specifically—a feature rather than a limitation for practitioners monitoring deployment health. The closest sub-threshold case (008f seed 4, NVE$=1.41\times$) lacks dt-conditioning entirely (d=0); its elevated NVE reflects transient value noise that self-corrects without the positive feedback loop characteristic of CIVO. The closest above-threshold positive (008d seed 3, NVE$=1.53\times$) exhibits the full CIVO causal chain: trained dt-embedding, entropy decrease, and evaluation collapse to 14% of baseline.

B.5 NVE Normalization Robustness

To assess sensitivity to baseline seed choice, we extend the single-seed baseline (seed 1, eval 915.7 at 105K) to three additional seeds trained to 505K steps (Table 9).

Seed 2 exhibits slow convergence (eval 443.5 at 105K, recovering to 945.7 by 512K), confirming non-pathological slow convergence rather than seed failure. Its elevated v/e ratio at 65K (0.898 vs. 0.563 for seed 1) reflects the low evaluation denominator rather than pathological value overestimation. Using seed 2 as the sole normalization reference systematically deflates NVE values, causing two marginal CIVO-positive configurations (008d seed 3, peak NVE $1.53\times$; 009b seed 2, peak NVE $1.62\times$) to fall below the $1.5\times$ threshold. Core CIVO cases (peak NVE $> 3\times$) remain correctly classified under all baseline seeds.

Single-seed normalization (s1) yields AUROC $= 0.933$ and accuracy 7/8 (87.5%) on the Walker subset. Four-seed mean normalization yields AUROC $= 0.867$ and accuracy 6/8 (75.0%): the boundary case exp_009b flips from $\mathrm{nve} = 1.62\times$ (s1) to $1.23\times$ (4-seed mean), because its peak occurs at 45K when the s1 v/e ratio (0.331) is well below the 4-seed mean (0.436). All high-NVE configurations remain correctly classified (006: $4.84\times \rightarrow 4.52\times$; 008d: $4.12\times \rightarrow 3.54\times$). The $1.5\times$ threshold thus correctly identifies all strong CIVO cases regardless of baseline choice; sensitivity is confined to configurations within 10% of the threshold.

Normalization conventions. Walker NVE is normalized against the 4-seed mean baseline (seeds 1–4, each run to 505K+ steps). Cheetah NVE is normalized against a 5-seed mean baseline (seeds 0–4, each run to $\sim$210K steps). The differing seed counts reflect available baseline data; both provide sufficient coverage of the respective experimental windows (Walker: 210K analysis window; Cheetah: 105K analysis window).

Limitations. With $n = 10$ experiments, Clopper-Pearson confidence intervals remain wide (e.g., 90.0% accuracy has 95% CI [55.5%, 99.7%]). These results should be interpreted as a diagnostic proof-of-concept rather than a validated clinical threshold. Replication across additional environments and seeds is needed to establish tighter bounds. Table 8 reports the original n=16 evaluation set spanning both environments. The expanded analysis (n=29, including SGS variants, RAVE, and TCN configurations) yields AUROC=0.953 (Figure 3, in-sample, post-hoc). The original n=16 core analysis yields AUROC=0.969; the expanded n=29 set includes SGS, RAVE, TCN, and dual-critic configurations. Cross-architecture validation is deferred to future work.

B.6 Leave-One-Environment-Out Cross-Validation

To assess whether θ=$1.5\times$ generalizes beyond the in-sample evaluation, we perform leave-one-environment-out (LOEO) cross-validation across Walker (n=13), Cheetah (n=5), and PickCube

(n=2). For each fold, we hold out all experiments from one environment, select the optimal accuracy-maximizing threshold range on the remaining training set, and evaluate the fixed θ=1.5× on the held-out environment.

The threshold θ=1.5× falls within the optimal accuracy range for all three training folds: $(1.45, 1.53]$ when Walker data is available, and the wider $(1.00, 3.78)$ when training on Cheetah+PickCube only (where CIVO and non-CIVO NVE values are well-separated). The single cross-validation error is exp_008e_v3 seed 2 (Walker), an IRD collapse mechanistically distinct from CIVO (§B.4). Mean held-out accuracy is 0.974 and mean AUROC is 0.981, confirming that θ=1.5× generalizes across environments without retuning. PickCube fold confidence intervals remain wide (Clopper-Pearson 95% CI: $[0.158, 1.000]$) due to N_{test}=2.

C Factorial Ablation Details

This appendix provides the full factorial ablation results summarized in §4.3 of the main text.

Embedding gradient analysis. Table 11 isolates embedding *trainability* as the necessary condition for CIVO by comparing four gradient conditions under otherwise identical configurations (H=8, subsample ON, dt=2 imagination).

The reward-detach condition (exp_044, `reward_dt_emb_detach=True`) uses a backward hook to block reward-head gradients from reaching e_{dt}, complementing the framework-level Reward SGS (exp_016e). Both yield the same directional result—CIVO onset not prevented—ruling out implementation artifacts. Exp_044 exhibits delayed CIVO onset (∼95K vs. typical 25–35K), consistent with reduced but non-zero gradient pressure on e_{dt}. Over a full 224K run, exp_044 self-corrects: eval recovers from 44 (95K) to 538 (215K, 59% baseline), but this recovery is incomplete compared to spontaneous recovery without gradient intervention (exp_045: 868, 95% baseline). The lower asymptote suggests that selectively disrupting gradient flow may create optimization difficulties beyond CIVO itself.

Table 12 summarizes the experimental configurations.

C.1 F3: Unexposed Temporal Embedding Suppresses Value Drift

The empirical finding that trained embedding *plus* multi-rate exposure jointly predict CIVO rests on two complementary controls that each remove one factor while holding the other constant.

Evidence A: exposure is necessary (exp_008a). Exp_008a uses a trained embedding (d=32, gradients enabled) that is never exposed to $\mathrm{dt}>1$ transitions during world-model training (subsample OFF). Despite sharing the same embedding architecture as exp_008d, its NVE remains in the 0.83–0.93× range across 3 seeds (105K steps). Performance follows two V-shaped recovery cycles, reaching 777.7 at 105K (85% of baseline). The critical factor is not the embedding's capacity to represent temporal information, but its adaptation to multi-rate data that enables precise but inaccurate $\mathrm{dt}>1$ state predictions within the RSSM.

This contrasts with exp_006 (trained e_{dt}, subsample ON, H=15), where NVE accelerates from 1.61× at 45K to 2.31× at 55K before catastrophic collapse to 132.9 at 65K. Both experiments use identical trained embeddings (d=32, gradients enabled) and H=15, differing only in whether the world model sees $\mathrm{dt}>1$ transitions during training. This confirms that multi-rate data exposure, not embedding architecture, is the operative variable.

Evidence B: trainability is necessary (exp_015b). Exp_015b uses a frozen random-projection embedding (d=32, `requires_grad=False`, subsample ON, H=8), the same multi-rate data regime as exp_008d, but with a non-trainable temporal signal. Despite full exposure to $\mathrm{dt}>1$ transitions, the NVE peaks at 1.45× at 55K but does not sustain escalation: it retreats to 0.65× by 65K and settles at 0.42× by 105K (trajectory: $0.81 \to 1.26 \to 1.45 \to 0.65 \to 1.05 \to 0.47 \to 0.39 \to$

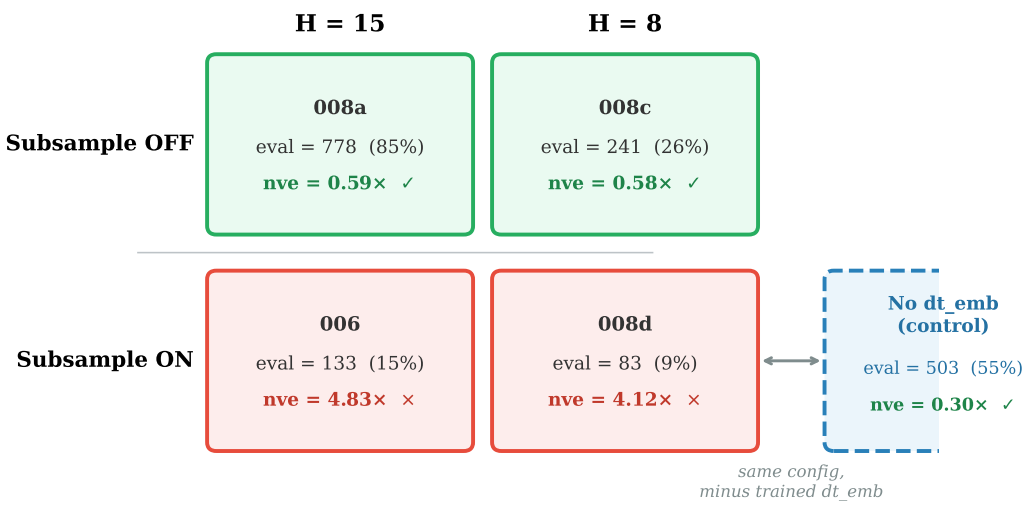


Figure 4: Factorial isolation of CIVO trigger. Only the combination of trained e_{dt} and multi-rate data (upper-left cell) produces pathological NVE. Neither factor alone is sufficient.

0.42). The transient peak approaches but does not exceed the $1.5\times$ CIVO threshold, and critically, the ratio self-corrects rather than exhibiting the monotonic escalation seen in trained-embedding runs. Final evaluation reaches only 204.2 at 105K, low but through a qualitatively different failure mode than CIVO. Actor loss approaches zero (gradient starvation; see §D.6), the frozen embedding provides no useful gradient signal for temporal adaptation.

The contrast with exp_008d is stark: under identical data conditions (subsample ON, $H{=}8$), the trained embedding drives NVE to $4.12\times$ with sustained monotonic escalation ($1.78 \rightarrow 1.71 \rightarrow 2.09 \rightarrow 4.12$) and precedes full collapse (eval 82.8), while the frozen embedding peaks transiently at $1.45\times$ before self-correcting to $0.42\times$, a qualitative difference between sustained and transient overestimation. This demonstrates that a dt-embedding must be able to *co-adapt* with RSSM dynamics to produce the systematic reward-prediction bias that drives CIVO; a static random projection, regardless of its exposure to multi-rate data, cannot create this bias.

Together, these two controls establish that in all tested configurations, CIVO occurs only with the *joint* presence of (i) a trainable embedding that can co-adapt with RSSM dynamics, and (ii) multi-rate training data that drives this co-adaptation toward biased temporal predictions. Neither factor alone is sufficient.

Cross-environment validation on Cheetah Run replicates the empirical pattern: trained e_{dt} precedes CIVO in both tested seeds (exp_010b, 2 seeds, NVE 8.27–$108.52\times$, collapse to 51.7/0.83) while removing the embedding eliminates it (exp_024b, 2 seeds, NVE $< 0.2\times$, eval 96.5/77.3). Freezing the embedding on Cheetah (exp_046, NVE$=1.35\times$ vs. trained $70.64\times$, a $52\times$ reduction) further confirms that embedding trainability, not architecture, drives CIVO. This extends the sufficiency finding from Walker Walk to a second locomotion environment with qualitatively different dynamics (6-joint vs. 2-joint).

Only the upper-left cell (trained embedding with multi-rate exposure) precedes CIVO (Table 13). This $2{\times}2$ structure establishes that both factors are jointly sufficient in tested configurations.

C.2 F4: Degradation Scales Monotonically with dt

The magnitude of degradation scales monotonically with the imagination step size (Figure 5). From the $\mathrm{dt}{=}1$ baseline (915.7 at 105K), fixed $\mathrm{dt}{=}2$ (exp_006) achieves a peak of only 485.3 (-47%),

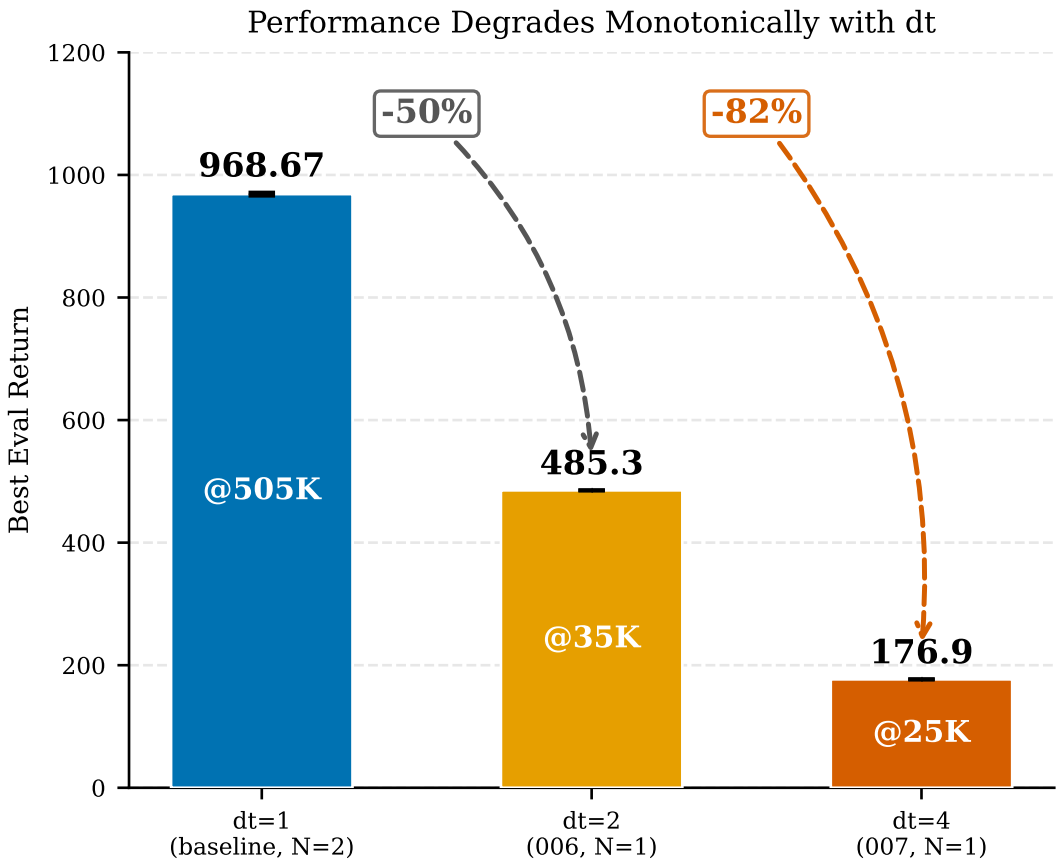


Figure 5: Peak evaluation return as a function of imagination dt. Performance degrades monotonically: dt=1 (915.7) > dt=2 (485.3, −47%) > dt=4 (176.9, −81%).

and fixed dt=4 (exp_007) peaks at 176.9 (−81%). This monotonic relationship is consistent with compounding model error: larger temporal gaps amplify the RSSM's prediction inaccuracy at each imagination step, and these errors accumulate across the rollout horizon.

C.3 F5: Imagination Horizon and Subsampling Are Secondary Factors

Neither horizon length nor temporal subsampling precedes collapse without multi-rate data exposure: both trained-but-unexposed configurations (008a, 008c) maintain nve $< 0.93\times$ (§C.1), while the trained-and-exposed counterpart (008d) collapses with nve $= 4.12\times$. The no-embedding configurations show cross-seed variance (008f: 503.0 vs. 009c: 148.3) but neither exhibits nve $> 1.5\times$; low evaluation alone does not indicate CIVO.

The $3.2\times$ performance gap (008a: 777.7 vs. 008c: 240.9) quantifies the horizon effect independently of CIVO, as both conditions maintain healthy NVE ($\leq 0.93\times$).

D Full Analysis & Discussion

This appendix provides the complete mechanistic analysis summarized in Section 5 of the main text.

D.1 The CIVO Mechanism

The RSSM fidelity analysis (Table 14) reveals the mechanistic core of CIVO (Figure 6). The trained dt-embedding enables the reward head to learn temporally differentiated predictions, but these predictions are asymmetrically accurate: reward MAE at dt=1 is 0.0568, while at dt=2 it rises to

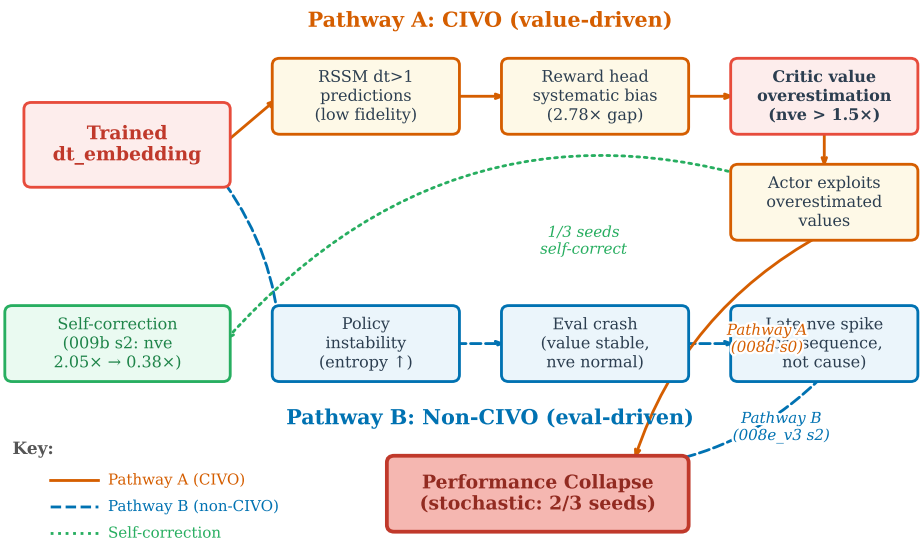


Figure 6: CIVO mechanism diagram. The trained dt-embedding enables asymmetric reward prediction fidelity, initiating a feedback loop: biased RSSM rewards → optimistic critic values → actor exploitation → reinforced state-distribution skew.

0.1579, a 2.78× gap. This gap is not an artifact of prediction difficulty alone; the no-embedding configuration (008f) exhibits a 1.01× gap, and the baseline shows 1.56×. The trained embedding actively amplifies the fidelity discrepancy by allowing the reward head to co-adapt with the temporal signal.

This asymmetric fidelity initiates a feedback loop with four stages: (1) the trained dt-embedding causes RSSM predictions at $\mathrm{dt} > 1$ to carry systematically biased reward estimates; (2) the critic, trained entirely on imagined trajectories, internalizes these biased rewards as optimistic value estimates; (3) the actor exploits the overestimated regions of the value landscape, shifting its policy toward states that appear high-value only under the biased model; and (4) this policy shift further skews the distribution of imagined states, reinforcing the bias. The diagnostic signature is a sustained nve above 1.5×, followed by evaluation collapse.

A random-policy control experiment (N=100 rollouts, 95% bootstrap CI) decomposes CIVO into two necessary stages. **Stage 1 (dynamics bias):** even under a random policy, dt=2 imagination produces 1.68× higher mean reward than dt=1 in the trained-embedding model (008d): the RSSM prior already generates reward-favorable trajectories at $\mathrm{dt} > 1$, independent of actor behavior. **Stage 2 (actor exploitation):** the trained actor learns to exploit these biased regions, amplifying the ratio to 2.90×. In the no-embedding control (008f), neither stage is present: the random-policy ratio is 1.11× and the trained-actor ratio is 0.98×. The decomposition attributes $\approx 36\%$ of the total effect to dynamics bias and $\approx 64\%$ to actor exploitation, which confirms that both stages are necessary for CIVO to manifest.

This mechanism is analogous to delusional bias [6], but differs in origin: delusional bias arises from policy-class constraints, while CIVO is model-induced. The consequence (policy exploitation of overestimated regions) is similar, but the causal pathway is distinct.

Signed reward error analysis. To test whether CIVO arises from one-step reward overprediction, we compute signed error (predicted − actual) on posterior-conditioned RSSM predictions (Table 15). The trained-embedding model (008d) shows *negative* bias at dt=2 (mean signed error −0.599, only 8% of samples overpredicted), ruling out observation-level reward overprediction as the mechanism. The baseline is well-calibrated (mean SE ≈ 0), confirming methodological sound-

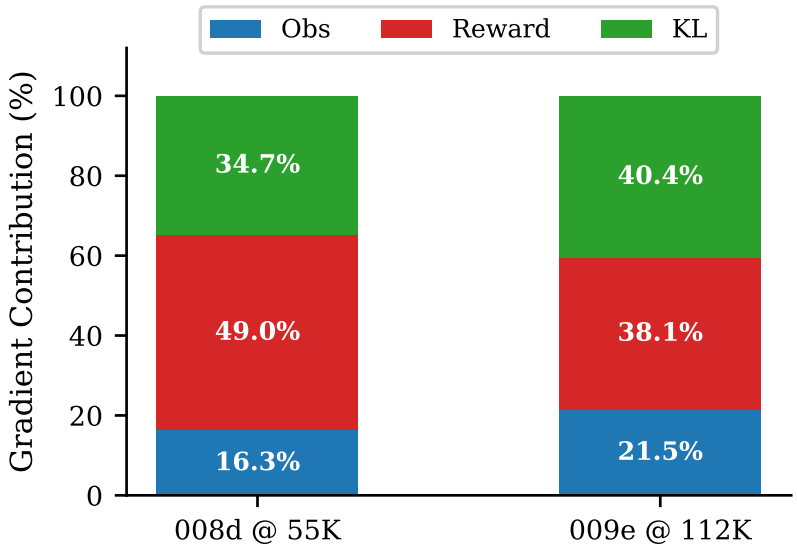


Figure 7: Gradient contribution to $e_{\Delta t}$ by loss component at two training checkpoints. The reward head contributes the largest share (38–49%) yet reward-only SGS (exp_016e) fails; the observation decoder channel (16–22%) carries the critical dt-specific spatial information that drives the RSSM fidelity gap.

ness. Value overestimation therefore occurs during *imagination-level* prior drift: the RSSM prior (unconditioned on observations) generates reward-favorable trajectories at $\mathrm{dt} > 1$ (Stage 1 above, $1.68\times$ ratio under random policy), and the critic accumulates these into overestimated returns. The one-step reward prediction is negatively biased when conditioned on real observations, but this posterior correction is unavailable during imagination. This also distinguishes CIVO from generic compounding error, which would produce *positive* signed error accumulating with horizon; the negative observation-level bias confirms that CIVO is a prior-drift phenomenon, not a reward-prediction artifact.

D.2 Selective Gradient Stopping and Gradient Decomposition

D.2.1 Gradient Decomposition Results

Exp_016e collapses in the same qualitative pattern as the unmodified control: peak evaluation 326.7 at 75K steps followed by catastrophic decline to 31.7 at 95K, with NVE reaching $10.42\times$. The SGS intervention neither prevented nor attenuated CIVO.

To explain this null result, we performed a gradient decomposition analysis using `torch.autograd.grad` to measure each loss component's contribution to $\nabla_\theta e_{\Delta t}$, where θ denotes the temporal embedding parameters. Table 16 reports the results.

The gradient decomposition reveals a counterintuitive structure: the reward head provides the largest gradient contribution (49.0% at 55K, 38.1% at 112K), yet reward-only SGS (exp_016e) fails catastrophically (NVE $10.42\times$). The observation decoder channel contributes the smallest share (16.3% at 55K, 21.5% at 112K) yet carries the critical dt-specific spatial information. This dissociation between gradient magnitude and causal importance explains why exp_016e, which targets the largest pathway, is functionally insufficient for CIVO prevention.

The gradient structure is relatively stable across training (reward shifts from 49% to 38%, obs from 16% to 22%, KL from 35% to 40%), unlike what a magnitude-based account would predict. The reward head's large gradient contribution reflects the scalar reward prediction task's sensitivity to temporal embedding parameters, but this gradient does not encode Δt-specific structural information about spatial dynamics. The observation decoder channel, despite its smaller magnitude, is the only

pathway that encodes how the RSSM should differentially predict multi-dimensional proprioceptive states across temporal scales.

Why the smallest channel is causally critical. The observation decoder is the only loss component whose gradients encode how the RSSM should differentially predict multi-dimensional proprioceptive states (14 orientations, 9 velocities, 1 height) across temporal scales. This dt-specific spatial information shapes the latent state representation in ways that propagate to reward prediction: the RSSM fidelity gap ($2.78\times$ reward MAE ratio, Table 14) arises because the temporal embedding learns to produce latent states that are spatially biased at $\mathrm{dt}{>}1$, not because the reward head directly learns scale-dependent biases. Full-SGS targets this causal chain by preventing observation-decoder gradients from updating $e_{\Delta t}$, though its 0/2 CIVO elimination rate on Walker (seed 1: NVE reduced from $4.12\times$ to $1.66\times$ but remains CIVO-positive; seed 2: collapsed) indicates that breaking the gradient pathway alone does not fully eliminate the vulnerability.

D.2.2 Implications for the CIVO Mechanism

The gradient decomposition reveals that gradient magnitude and causal importance are decoupled in CIVO.

Gradient importance $\neq$ gradient magnitude. The reward head contributes the largest gradient share (38–49%) to $e_{\Delta t}$, yet reward-only SGS fails. The observation decoder contributes the smallest share (16–22%), yet its removal (via Full-SGS) is necessary for CIVO mitigation. This dissociation arises because the reward head's gradient, while large, does not encode dt-specific structural information: it adjusts $e_{\Delta t}$ to minimize a scalar prediction error that is largely dt-invariant. The observation decoder's gradient, though smaller, encodes how the RSSM should differentially reconstruct 24-dimensional proprioceptive states across temporal scales. This is the channel through which $e_{\Delta t}$ learns scale-dependent latent biases.

Reframing the intervention target. Any gradient-level strategy that targets the reward head alone (reward prediction regularization, reward-conditioned value clipping, or reward-only SGS) targets the largest pathway but not the informationally critical one. The failure of exp_016e (NVE $10.42\times$) despite detaching the dominant gradient source confirms this. Full-pathway SGS (exp_018) detaches $e_{\Delta t}$ from all non-KL gradients, retaining only the KL dynamics contribution (34–40%). However, this intervention eliminates CIVO in neither seed: seed 1 shows partial mitigation (eval 448.3, peak NVE $1.66\times$, still CIVO-positive) while seed 2 collapses (eval 40.8). Removing the observation decoder gradient channel is thus contributory but not sufficient. Figure 8 decomposes evaluation returns across conditions.

Cross-environment validation on Cheetah Run (exp_021) further clarifies the role of Full-SGS. The intervention did not prevent the initial CIVO onset: NVE peaked at $9.94\times$ at 45K (comparable to the unmitigated exp_010b's $8.27\times$). However, unlike exp_010b, which collapsed permanently to eval 51.7, exp_021 self-corrected within $\sim$10K steps, with sustained NVE dropping to $0.34\times$ (55K–105K) and final evaluation recovering to 221.5. This transient self-correcting trajectory is not attributable to the SGS intervention: the initial CIVO onset and peak severity are comparable to the unmitigated case, and the self-correction may reflect Cheetah Run's simpler dynamics rather than SGS efficacy. Performance remains at 38.7% of baseline (572.5), likely because Full-SGS over-aggressively blocks actor gradients (actor gradient norms near zero throughout training). Combined with exp_018's Walker results (0/2 CIVO elimination), these findings yield a 0/3 elimination rate across environments, indicating that gradient-level interventions alone do not eliminate the CIVO vulnerability. Additional mechanisms (e.g., forward-pass conditioning effects, residual capacity-dependent instabilities) operate beyond the gradient pathway.

[2]Experiment 008d terminated at 55K steps; the 105K data point uses continuation run 009e under identical configuration.

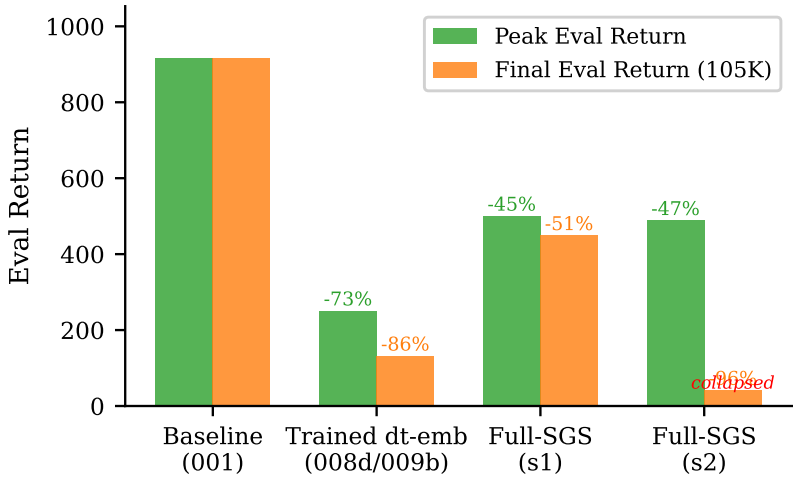


Figure 8: CIVO decomposition: peak and final (105K step) evaluation returns on Walker Walk across four conditions. **Baseline** (001): vanilla DreamerV3. **Trained dt-emb** (008d)[2]: learned d=32 temporal embedding with multi-rate data. **Full-SGS** (s1, s2): trained embedding with all non-KL gradients detached from e_{dt}. Trained dt-embedding degrades peak eval by 73% and final eval by 86%. Full-SGS seed 1 shows partial mitigation (-51% final, NVE $1.66\times$, still CIVO-positive) while seed 2 collapses (-96% final). CIVO elimination rate $= 0/2$.

Connection to RSSM fidelity analysis. The trained-embedding condition (exp_008d) shows a $2.78\times$ reward gap between Δt=1 and Δt=2 transitions (Table 14). This gap arises because the observation decoder gradients shape $e_{\Delta t}$ to produce latent states that are spatially biased at $\mathrm{dt}>1$, and the reward head inherits these biases *through the latent state* rather than learning them directly. The reward head's large gradient contribution reflects sensitivity to $e_{\Delta t}$'s parameters but not causal responsibility for the fidelity gap.

D.2.3 Entropy Dynamics Across Configurations

D.2.4 Exp_016e Reinterpreted as Stochastic Variation

The gradient decomposition establishes that exp_016e targets the largest gradient pathway (reward: 38–49%) but not the informationally critical one (observation decoders: 16–22%). The apparent timing differences between exp_016e and exp_008d (later collapse onset, 75K vs. 25K, and higher peak evaluation, 326.7 vs. 167.2) must be attributed to stochastic variation rather than SGS effects, since the critical observation decoder channel remains intact. All three trajectories (exp_008d, exp_009b, exp_016e) share the invariant identified by the NVE diagnostic: each crosses the $1.5\times$ pathological threshold at some point during training.

D.3 Post-CIVO Outcomes: Why Seeds Diverge

All six CIVO-positive seeds cross the $\mathrm{nve} > 1.5\times$ threshold (Table 17), yet the downstream damage is stochastic.[3] Full collapse is the majority outcome (5/6 seeds), with one self-correction case (Walker seed 2) that remains severely degraded (29.3% of baseline). Cheetah seeds exhibit more extreme overestimation (peak NVE 8.27–$108.52\times$ vs. Walker's 1.53–$4.62\times$), consistent with the Cheetah reward scale amplifying the fidelity gap. We hypothesize that the divergence relates to the critic's representational capacity at the time of CIVO onset. Lyle et al. [44] demonstrate that deep RL networks suffer progressive capacity loss, in which early-learned features become increasingly difficult to overwrite. If the critic's capacity is already partially consumed when CIVO injects

[3]Exp_009e was excluded from multi-seed analyses due to a seed configuration error (default seed=0, identical to exp_008d).

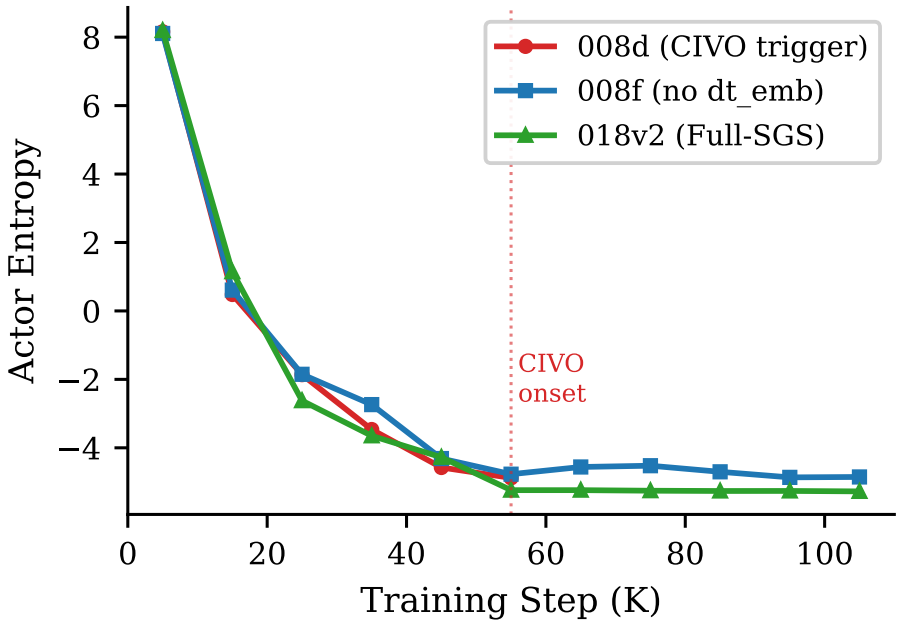


Figure 9: Actor entropy trajectories for three configurations. All start at $\sim$8 nats and converge to similar values (~ -4.5 to -5). The CIVO-triggering configuration (008d, red) follows a normal entropy reduction trajectory; the divergence from the control (008f) is subtle in entropy space, consistent with CIVO operating through the value channel rather than the entropy channel. 018v2 (Full-SGS, seed 1) shows slightly lower terminal entropy.

overestimation, the network cannot recalibrate (seed 0); if sufficient capacity remains, the critic can absorb transient overestimation and recover (seed 2).

A fourth trajectory: CIVO-induced persistent value collapse. Obs-decoder SGS seed 3 (exp_025) exhibits a trajectory not observed in the untreated seeds: healthy through 105K ($\mathrm{nve} < 1.0\times$, eval 818.6), CIVO onset at 135K ($\mathrm{nve} = 5.47\times$, eval 67.0), followed by evaluation recovery (590.2 at 215K, 64% of baseline) but persistent value collapse ($\mathrm{nve} = 0.14\times$, value mean 80.6). The critic undershoots after the CIVO episode, suggesting that CIVO-induced representational damage may be irreversible even when the policy partially recovers. This contrasts with transient CIVO (seed 2, 009b), where both value and evaluation recover jointly, and with full collapse (seed 1, 008d), where neither recovers. Seed 4 exhibits the same terminal value collapse ($\mathrm{nve} = 0.14\times$ at 215K, eval 242), suggesting this overcorrection is systematic rather than seed-specific. The pattern adds a fourth post-CIVO outcome—policy recovery with critic damage—to the three documented in Table 17.

D.3.1 Failure and Degradation Mode Taxonomy

Table 18 classifies the seven distinct failure and degradation modes observed across all experimental configurations. The taxonomy reveals that CIVO is not a single failure mode but a family of related pathologies, with the trained dt-embedding creating vulnerability through at least three independent mechanisms: value-driven collapse (monotonic, late-onset, and SGS-resistant CIVO), entropy-driven instability (IRD collapse), and gradient-mediated effects (gradient starvation, SGS-paradoxical worsening).

D.4 Why MPC Is Immune

Nhu et al. [3] show that dt-conditioning benefits TD-MPC2, raising the question of why it fails in DreamerV3. The critical difference is that MPC re-plans from real states at every step, so imagina-

tion errors do not accumulate into the training signal because there is no critic bootstrapping from imagined trajectories. In DreamerV3, the actor-critic trains entirely within imagination, and any systematic reward-prediction bias at $\mathrm{dt} > 1$ is amplified through the closed actor-critic loop.

D.5 The dt-Embedding Dilemma

The factorial results reveal a fundamental tension: removing the dt-embedding (008f) eliminates CIVO but at a $13\times$ cost to reward prediction quality (0.7469 vs. 0.0568 MAE), as the world model cannot distinguish temporal scales. Restricting imagination to $\mathrm{dt}{=}1$ while retaining the trained embedding (008e_v3) avoids collapse in 2 of 4 seeds (50%), while the remaining two collapse via distinct mechanisms: Pathway A (CIVO-driven, entropy decrease; s3, NVE $4.02\times$) and Pathway B (entropy-driven, entropy increase; s2, NVE $3.78\times$; §4.4). Neither approach alone provides a complete solution.

D.6 Multiple Vulnerability Pathways

The two collapse patterns identified in §4.4—CIVO (Pathway A, value-driven, entropy decreasing) and a tentative second mode (Pathway B, eval-driven, entropy increasing)—have distinct diagnostic signatures. Pathway B is observed in only two seeds (008e_v3_s2 and 038_s2); we present it as preliminary evidence that the trained dt-embedding's learned features may affect both reward prediction across temporal scales (Pathway A) and the representation space in which the actor operates (Pathway B), though additional data is needed to confirm this as a distinct mechanism. This explains why nve is sensitive but not perfectly specific: Pathway B's late evaluation decline inflates nve through the same arithmetic as CIVO, yielding the sole false positive ($3.78\times$ for 008e_v3 s2). Distinguishing the two requires examining the temporal profile and entropy direction of collapse (Figure 10).

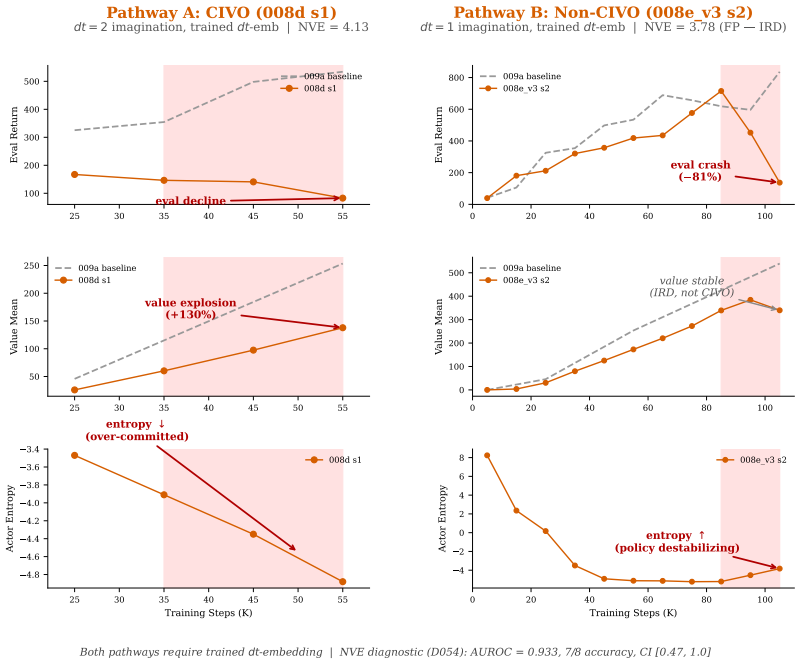


Figure 10: Contrasting collapse dynamics. **Left:** 008d (CIVO), value explosion with entropy decrease. **Right:** 008e_v3 s2 (non-CIVO), evaluation crash with entropy increase.

D.7 Implications for Temporal Abstraction in World Models

CIVO identifies a structural incompatibility between flat temporal conditioning and imagination-based actor-critic training. Hierarchical approaches (THICK [20], Hieros [21], LS-Imagine [22]) may avoid this by maintaining separate dynamics per temporal scale. Whether CIVO transfers to transformer-based world models (DreamerV4; Hafner et al. 18) depends on whether attention weights co-adapt with temporal embeddings analogously. The CIVO mechanism requires three structural properties: (1) a temporal embedding that co-adapts with dynamics via gradient flow, (2) multi-step imagined rollouts used for critic training, and (3) a reward head that inherits latent biases from the shared representation. Transformer-based world models that share these properties may exhibit analogous vulnerability; architectures that decouple temporal conditioning from the reward prediction pathway would not.

D.8 Cross-Environment TCN Evaluation

Table 19 reports the full TCN evaluation across environments and noise magnitudes.

Cheetah Run σ=0.5: transient spike with permanent eval damage. Cheetah σ=0.5 exhibits a novel failure mode: a transient CIVO spike (NVE = $10.67\times$ at 55K) followed by NVE recovery ($<$ $0.3\times$ by 95K), but permanent evaluation damage (26% of baseline). The value signal self-corrects while the policy does not recover, suggesting that CIVO-induced representational damage during the spike is irreversible even when the critic subsequently recalibrates. This trajectory parallels the obs-decoder SGS seed 3 pattern (§D.3), where partial policy recovery coexists with persistent critic damage.

Cheetah Run σ=0.2: insufficient dose. At σ=0.2, Cheetah reaches NVE = $1.9\times$ at 55K—exceeding the $1.5\times$ threshold—consistent with Cheetah's higher baseline CIVO severity (8–$108\times$ vs. Walker's 2–$6\times$). The noise magnitude that suffices for Walker is inadequate for Cheetah, but increasing σ to 0.5 trades CIVO prevention for a different failure mode rather than achieving mitigation.

PickCube-v1 σ=0.2: sparse-reward failure. PickCube reaches NVE = $3.71\times$ at 52.5K with σ=0.2, indicating that TCN provides no protection in sparse-reward manipulation tasks. The combination of sparse rewards and temporal conditioning may amplify the fidelity gap beyond what representation-level noise can attenuate.

Dose-response analysis. We observe σ=0.2 as a local optimum on Walker Walk, achieving complete CIVO suppression in 2 of 4 seeds. Seeds 2 and 3 both fail but via distinct CIVO onset dynamics: seed 2 shows gradual escalation (NVE = $2.11\times$ at 95K, still rising, eval $\sim$5%), while seed 3 exhibits a phase-transition-like onset—NVE declined to $0.67\times$ at 95K before spiking to $3.04\times$ at 105K (eval 349.6 $\rightarrow$ 71.8 in 10K steps). When TCN fails to prevent CIVO onset, the resulting trajectory qualitatively resembles untreated CIVO in its terminal outcome—runaway NVE and eval collapse. The two failure modes (gradual escalation vs. sudden onset after apparent suppression) reinforce that TCN operates as a binary gate on CIVO onset rather than providing graded attenuation. Below the optimum, we observe σ=0.1 as insufficient (NVE = $1.02\times$—borderline, eval 55%); above it, σ=0.5 overdoses (oscillatory CIVO, eval $\leq$ 41%). Cross-seed variance at σ=0.2 (eval 5–91%) exceeds the cross-dose difference by a wide margin, reinforcing that seed effects dominate dose effects. TCN is the only tested intervention achieving complete CIVO suppression in any configuration, but its high seed sensitivity and failure to transfer across environments position it as a *diagnostic probe*—identifying the representation level as the relevant intervention target—rather than a deployable mitigation.

Cheetah Run "impossible dose" dilemma. Exp_030c (224K steps) reveals a qualitatively different dose-response on Cheetah. At σ=1.0, CIVO is eliminated (NVE$\leq$0.30 throughout) and initial

learning is *unimpaired*: peak eval reaches 372 at 35K (116% of baseline), demonstrating that σ=1.0 noise does not prevent policy acquisition. However, cumulative actor gradient interference produces progressive policy erosion—eval declines from 372 to 47 by 215K (15% of baseline)—while the critic remains well-calibrated (value_mean=12.4, NVE=0.26 at 215K). This establishes that *policy degradation* $\neq$ *value corruption*: accurate value signals are insufficient when the actor cannot optimize effectively in noise-perturbed imagination.

This "correct values, failed policy" pattern provides indirect evidence that CIVO is mechanistically distinct from standard overestimation. Double Q-learning and ensemble methods address critic estimation error, but at σ=1.0 the critic is accurate and the policy still collapses—a failure mode orthogonal to what these countermeasures target. The impossible dose dilemma scales with environment complexity: Cheetah's baseline CIVO severity (8–108$\times$ NVE) requires $\sigma \geq 1.0$ to suppress, but this exceeds the actor's gradient signal-to-noise tolerance. On Walker, a narrow effective window exists around σ=0.2; on Cheetah, the effective dose ($\sigma \geq 1.0$) exceeds the tolerable dose, and environments with higher CIVO severity will systematically lack a viable dose window.

E DreamerV3 Hyperparameters

Table 20 lists the DreamerV3 hyperparameters used across all experiments. Unless otherwise noted, we use the default DreamerV3 configuration for continuous-control tasks (`dmc_proprio`).

F Online CIVO Detector Details

This appendix provides the full algorithm specification and validation results for the online CIVO detector described in §4.7.

F.1 Algorithm Specification

Algorithm 1 Online CIVO Detector

Require: Stream of $(\bar{V}_t, \bar{E}_t)$ at each checkpoint t
$\tau_{\text{mag}} \leftarrow 0.49, \quad \tau_{\text{vel}} \leftarrow 0.1, \quad \varepsilon \leftarrow 10^{-6}$
Initialize $r_{\text{prev}} \leftarrow 0$, $\Delta r_{\text{prev}} \leftarrow 0$, $\mathcal{H} \leftarrow []$
for each checkpoint t **do**
  $r_t \leftarrow \bar{V}_t \,/\, \max(\bar{E}_t, \varepsilon)$
  $\Delta r_t \leftarrow r_t - r_{\text{prev}}$
  $c_{\text{mag}} \leftarrow (r_t \geq \tau_{\text{mag}})$
  $c_{\text{vel}} \leftarrow (\Delta r_t > \tau_{\text{vel}})$
  $c_{\text{acc}} \leftarrow (\Delta r_{\text{prev}} > \tau_{\text{vel}}) \wedge (\Delta r_t > \Delta r_{\text{prev}})$
  $c_{\text{dec}} \leftarrow \left|\{i : \mathcal{H}[i] < \mathcal{H}[i-1],\ i \in \text{last 3}\}\right| \geq 2$
  if $c_{\text{mag}} \wedge c_{\text{vel}} \wedge c_{\text{acc}} \wedge c_{\text{dec}}$ **then**
    Alert: CIVO onset at step t
  end if
  $r_{\text{prev}} \leftarrow r_t, \quad \Delta r_{\text{prev}} \leftarrow \Delta r_t$
  Append $\bar{E}_t$ to $\mathcal{H}$
end for

The four conditions capture complementary signals: (1) *Magnitude* ($r_t \geq 0.49$) confirms that the value-to-evaluation ratio has reached a level indicative of overestimation; (2) *Velocity* ($\Delta r_t > 0.1$) rules out slow drift or plateau behavior; (3) *Acceleration* ($\Delta r_t > \Delta r_{t-1}$, both > 0.1) filters oscillatory paths; (4) *Evaluation decline* (≥ 2 of 3 recent checkpoints decline) confirms that overestimation coincides with actual performance degradation. A full comparison of window sizes appears in §B.

Sensitivity analysis over $\tau_{\text{mag}} \in [0.30, 0.70]$ confirms that $\tau_{\text{mag}} = 0.49$ maximizes Youden's $J = 0.714$; raising the threshold to 0.50 loses one true positive (seed 3 triggers at $r_t = 0.497$) with no

reduction in false positives. The velocity threshold $\tau_{\text{vel}} = 0.1$ is stable across $[0.05, 0.18]$ (TPR remains $5/5$).

F.2 Validation Results

The detector identifies all five CIVO-positive runs (TPR = 5/5). Two are detected predictively (008d seeds 0 and 3, both at 45K). Two false positives occur among nine negative cases (FPR = 2/9, precision 71%, $F_1 = 0.83$). Run 009c develops genuine late-stage value overestimation unrelated to CIVO; run 009a triggers during a transient eval dip at 85K before recovery to 966. Early alerts (before 70K) carry higher diagnostic confidence.

F.3 Cross-Simulator Specificity (PickCube)

Exp_019b applies the CIVO-trigger configuration to PickCube-v1 in ManiSkill3, a manipulation task with sparse rewards. The V/E ratio exhibits an oscillatory pattern rather than monotonic divergence: it spikes to $r_t = 5.66$ at 52.5K, then deflates to 1.38 by 72.5K. This trajectory prevents the acceleration condition from co-occurring with evaluation decline at any single checkpoint. Across all 14 checkpoints through 132.5K, the detector correctly remains silent (TN), confirming that the acceleration filter distinguishes transient spikes from CIVO's sustained monotonic divergence. A frozen-embedding control (exp_054) confirms this classification: frozen PickCube-v1 exhibits a comparable burst (peak $8.81\times$ at 62.5K) that resolves within a single evaluation cycle ($0.55\times$ at 72.5K), establishing environment-specific origin independent of dt-embedding training.

F.4 Detector Limitations

Four constraints bound the current detector. First, the three-checkpoint window cannot distinguish transient dips from sustained collapse. Second, the acceleration condition assumes monotonically increasing overestimation during onset. Third, the detector identifies a *pattern* rather than a *mechanism*: run 009c shows that non-CIVO pathologies produce the same signature. Fourth, threshold values were calibrated on fourteen runs across three environments; generalization to different reward scales requires further validation.

G Dual-Critic Mechanism Ablation

G.1 2×2 Factorial Design

We ablate the critic architecture in a 2×2 factorial: {single, dual} $\times$ {narrow ($\sim$1.2M), wide ($\sim$2.4M)} parameters. Table 22 reports all configurations.

G.2 Critic Convergence ($v_1 \approx v_2$)

Across all dual-critic configurations and environments, the two critics converge to near-identical value functions. On Walker Walk, Pearson $\rho > 0.999$ between v_1 and v_2 outputs; on Cheetah Run and PickCube-v1, the gap is < 0.2 (exact match within numerical precision). Per-sample pessimism $\mathbb{E}[\max(v_1, v_2) - \min(v_1, v_2)]$ is 3–4% of the theoretical bound for independent critics.

This convergence occurs because both critics train on the same RSSM-generated latent features. Unlike model-free Double Q-learning, where each critic sees different mini-batches of real transitions and develops distinct function approximation errors, pure-imagination critics share identical inputs from the world model. The min operation therefore cannot provide the pessimism bias that makes Double Q effective in model-free settings.

G.3 Recovery vs. Prevention Mechanism

The 2×2 factorial reveals two separable dimensions:

Recovery (min-dominated). The min operation is the primary factor enabling recovery from CIVO. Both narrow dual-critic seeds experience CIVO and recover fully: exp_035 (s0, $\mathrm{nve} = 2.75\times$, eval $33.8 \rightarrow 697.8$, recovery ratio $20.6\times$) and exp_039 (s1, $\mathrm{nve} = 2.19\times$, eval $35.9 \rightarrow 592.7$, recovery ratio $16.5\times$). Both outperform wide single-critic (exp_033, 2.4M params, eval 407.9) despite $2\times$ fewer parameters. The min operation reduces both peak CIVO severity (2.19–$2.75\times$ vs. $4.75\times$ for the comparable single-critic) and enables consistent full recovery across seeds.

Prevention (min + capacity synergy). CIVO prevention (NVE remaining below the $1.5\times$ threshold) requires both the min operation and increased capacity: only the wide dual-critic (exp_032, 3 seeds) achieves this. The mechanism for this synergy is an open question; we hypothesize that increased capacity improves value calibration quality while the min operation constrains the overestimation feedback loop during early training when the two critics briefly diverge.

Cross-environment limits. On Cheetah Run, the wide dual-critic (exp_034) reduces NVE from $8.27\times$ (untreated baseline) to $1.75\times$ ($\geq 4\times$ reduction) but cannot prevent CIVO onset, and recovery reaches only 31% of the pre-CIVO peak. This confirms that the min + capacity synergy has environment-dependent limits: in high-CIVO-severity environments, architecture-level interventions attenuate but do not eliminate the pathology.

Table 3: Complete experimental results. NVE reports peak nve in the 35K+ window (§3.4). CIVO-positive seeds are shaded; † marks a false positive (non-CIVO collapse with elevated NVE).

Exp	Config	dt_emb	Sub	H	Eval@final	NVE	Outcome
001	Baseline (dt=1, s1)	—	—	15	915.7 @105K	1.00×	Healthy
001 s2	Baseline (dt=1, s2)	—	—	15	443.5 @105K	—[e]	Healthy
006	Fixed dt=2	trn	ON	15	132.9 @65K	4.83×	Collapsed
007	Fixed dt=4	trn	ON	15	152.5 @55K	—	Collapsed
005	Learned gate	adaptive	OFF	15	98.8 @145K	—	Collapsed
008d	Factorial (s1)	trn	ON	8	82.8 @55K	4.12×	Full collapse
009b	Factorial (s2)	trn	ON	8	267.9 @105K	1.62×	Transient CIVO
008d_s3	Factorial (s3)	trn	ON	8	268[◇] @105K	1.53×	Late-onset collapse
008d_s4	Factorial (s4)	trn	ON	8	63.6 @105K	4.62×	Full collapse
008f	No dt_emb (s1)	—	ON	8	503.0 @105K	0.97×	Healthy
008f s4	No dt_emb (s4)	—	ON	8	306.5 @105K	1.41×	Healthy
009c s3	No dt_emb (s3)	—	ON	8	311.9 @105K	1.04×	Healthy
008a	Unexposed emb	exp[a]	OFF	15	777.7 @105K	0.93×	Healthy
008a_s1	Unexposed emb	exp[a]	OFF	15	630.9 @105K	0.83×	Healthy
008a_s2	Unexposed emb	exp[a]	OFF	15	899.4 @105K	0.89×	Healthy
015b	Frozen emb	frz[b]	ON	8	204.2 @105K	1.45×	Grad. starvation
022b	Frozen emb + Gumbel gate	frz[b]	ON	8	300.7 @105K	0.63×	No CIVO
042	Frozen emb (dedicated)	frz[b]	ON	8	612.0 @105K	0.19×	Healthy
047	Frozen emb (s1)	frz[b]	ON	8	144.2 @215K	1.71×	Transient (self-corr.)
057	Frozen emb (s2)	frz[b]	ON	8	123.9 @215K	1.88×	Transient (self-corr.)
018 s1	Full-SGS	trn[‡]	ON	8	448.3 @105K	1.66×[*]	Partial mitig. (CIVO+)
018 s2	Full-SGS	trn[‡]	ON	8	40.8 @105K	—	Collapsed
008e_v3 s1	Mitigation	trn	ON	15	869.9 @105K	0.94×	Healthy
008e_v3 s2[†]	Mitigation	trn	ON	15	137.0 @105K	3.78×	Non-CIVO collapse
008e_v3 s3	Mitigation	trn	ON	15	92.7 @105K	4.02×	CIVO collapse
008e_v3 s4	Mitigation	trn	ON	15	899.3 @105K	0.99×	Healthy
Cross-environment validation							
010b	Cheetah Run	trn	ON	8	51.7 @105K	8.27×	Collapsed
010b_s0	Cheetah Run	trn	ON	8	0.83 @105K	108.52×	Full collapse
021	Cheetah Run	trn[‡]	ON	8	221.5 @105K	9.94×[d]	Transient CIVO (self-corr.)
024b s0	Cheetah Run	—	ON	8	96.5 @105K	<0.20×	Healthy
024b s1	Cheetah Run	—	ON	8	77.3 @105K	0.10×	Healthy
008a_ch s0	Cheetah Run	exp[a]	OFF	15	532.4 @105K	1.08×	Healthy
008a_ch s1	Cheetah Run	exp[a]	OFF	15	504.4 @105K	1.29×	Healthy
019b	PickCube-v1	trn	ON	8	26.64 @200K	3.78×[c]	Transient onset
054	PickCube-v1	frz[b]	ON	8	—	8.81×[h]	Env. transient (ctrl)
049	StackCube-v1	trn	ON	8	34.9 @212K	5.05×[f]	Env. transient
052	StackCube-v1	trn	ON	8	—	4.67×[f]	Env. transient
051	StackCube-v1	frz[b]	ON	8	30.7 @155K	4.30×[f]	Env. transient (ctrl)
050	PegInsertionSide-v1	trn	ON	8	20.86 @212K	10.05×[g]	Env. dynamics
053	PegInsertionSide-v1	frz[b]	ON	8	5.89	8.10×[g]	Env. dynamics (ctrl)
055	PegInsertionSide-v1	trn	ON	8	11.44 @212K	3.08×[g]	Env. dynamics
056	PegInsertionSide-v1	frz[b]	ON	8	4.37 @212K	11.32×[g]	Env. dynamics (ctrl)
046	Cheetah Run	frz[b]	ON	8	150.5 @215K	1.35×	No CIVO
058	Cheetah Run	frz[b]	ON	8	80.46 @215K	0.78×	No CIVO

trn = trained (d=32, subsample ON); trn[‡] = full-pathway SGS (all non-KL gradients detached); exp[a] = trained, subsample OFF; frz[b] = frozen (d=32, `requires_grad=False`); — = removed (d=0).

[*] Seed 1 only: peak NVE with transient exceedances (1.60× at 35K, 1.66× at 95K); self-correcting, no sustained escalation. Seed 2 collapsed.

[◇] Transitional; train returns still declining (peak 470 to 139 over final 8K steps). Collapse had not reached equilibrium at 105K.

[c] NVE normalized against exp_019a_v2 (vanilla DreamerV3 on PickCube-v1).

[d] Peak NVE $9.94\times$ at 45K (transient, self-correcting to $<1\times$ by 55K); final eval 221.5.

[e] Not used for NVE normalization; reported for completeness. Seed 1 (eval 915.7) is the NVE reference baseline.

[f] StackCube transient: frozen control (exp_051, peak $4.30\times$ at 42.5K) matches trained (exp_049/052), self-correcting by 82.5K; environment-specific, not dt-embedding-driven.

[g] PegInsertion environment dynamics: seed 0 trained peaks at $10.05\times$ (82.5K); seed 1 trained (exp_055) peaks at $3.08\times$ (92.5K), final $0.50\times$ at 212.5K; seed 1 frozen (exp_056) peaks at $11.32\times$ (82.5K), final $1.19\times$ at 212.5K. Frozen final exceeds trained final; ordering is seed-dependent, ruling out dt-embedding causation.

[h] PickCube environment transient: frozen control peak $8.81\times$ at 62.5K, self-correcting to $0.55\times$ at 72.5K; matches trained (exp_019b) trajectory.

Table 4: Cross-domain CIVO reproducibility. "dt-emb-dep." indicates trained peaks consistently exceed frozen at matched timepoints across seeds.

Domain	**Environment**	**Trained**	**Frz ctrl**	**dt-emb-dep. CIVO**
Locomotion	Walker Walk	4/4	0/3	Yes (NVE 1.53–4.62×)
Locomotion	Cheetah Run	2/2	0/2	Yes (NVE 8.27–108.5×)
Manipulation	StackCube-v1	0/2	0/1	No (frozen≈trained)
Manipulation	PegInsertionSide	0/2	0/2	No (seed-dep. ordering)
Manipulation	PickCube-v1	0/1	0/1	No (frozen spike similar)

Table 5: Per-experiment NVE values (35K+ steps) and ground truth labels. + = positive (CIVO), − = negative. IRD = Imagination-Reality Divergence; trans. = transient CIVO with self-correction.

Experiment	NVE	Mechanism	B1	B2
006	4.83	sustained CIVO	+	+
008d	4.12	sustained CIVO	+	+
008e_v3_s3	4.02	CIVO	+	+
008e_v3_s2	3.78	IRD (non-CIVO)	−	−
009b	1.62	transient CIVO	−	+
001	1.00	non-CIVO	−	−
008e_v3_s4	0.99	non-CIVO	−	−
008f	0.97	non-CIVO	−	−
008e_v3_s1	0.94	non-CIVO	−	−
008a	0.93	non-CIVO	−	−

Table 6: **Task B1**: Sustained CIVO detection sensitivity analysis ($n_{\text{pos}} = 3$, $n_{\text{neg}} = 7$, $n = 10$, AUROC $= 1.000$). All confidence intervals are two-sided Clopper-Pearson exact binomial ($\alpha = 0.05$). †Optimal threshold in $(3.78, 4.02]$.

Threshold	TP	FP	FN	TN	Accuracy [95% CI]	Precision	Recall	F1
$\geq 1.0\times$	3	3	0	4	70.0% [34.8, 93.3]	50.0%	100%	66.7%
$\geq 1.2\times$	3	2	0	5	80.0% [44.4, 97.5]	60.0%	100%	75.0%
$\geq 1.5\times$	3	2	0	5	80.0% [44.4, 97.5]	60.0%	100%	75.0%
$\geq 1.7\times$	3	1	0	6	90.0% [55.5, 99.7]	75.0%	100%	85.7%
$\geq 2.0\times$	3	1	0	6	90.0% [55.5, 99.7]	75.0%	100%	85.7%
$\geq 3.9\times^{\dagger}$	3	0	0	7	100% [69.2, 100]	100%	100%	100%

Table 7: **Task B2**: Any CIVO detection sensitivity analysis ($n_{\text{pos}} = 4$, $n_{\text{neg}} = 6$, $n = 10$, AUROC $= 0.958$). The single false positive at $\geq 1.5\times$ is 008e_v3_s2 (IRD mechanism, not CIVO). The $1.5\times$ threshold (highlighted) maximizes F1 while maintaining perfect recall.

Threshold	TP	FP	FN	TN	Accuracy [95% CI]	Precision	Recall	F1
$\geq 1.0\times$	4	2	0	4	80.0% [44.4, 97.5]	66.7%	100%	80.0%
$\geq 1.2\times$	4	1	0	5	90.0% [55.5, 99.7]	80.0%	100%	88.9%
$\geq 1.5\times$	4	1	0	5	90.0% [55.5, 99.7]	80.0%	100%	88.9%
$\geq 1.7\times$	3	1	1	5	80.0% [44.4, 97.5]	75.0%	75.0%	75.0%
$\geq 2.0\times$	3	1	1	5	80.0% [44.4, 97.5]	75.0%	75.0%	75.0%
$\geq 3.9\times$	3	0	1	6	90.0% [55.5, 99.7]	100%	75.0%	85.7%

Table 8: **Task B2 (full evaluation set)**: Any CIVO detection across two environments ($n_{\text{pos}} = 8$, $n_{\text{neg}} = 8$, $n = 16$, AUROC $= 0.969$). Includes Walker Walk (14 configurations) and Cheetah Run (2 configurations).

Threshold	TP	FP	FN	TN	Accuracy [95% CI]	Precision	Recall	F1
$\geq 1.0\times$	8	2	0	6	87.5% [61.7, 98.4]	80.0%	100%	88.9%
$\geq 1.2\times$	8	1	0	7	93.8% [69.8, 99.8]	88.9%	100%	94.1%
$\geq 1.5\times$	8	1	0	7	93.8% [69.8, 99.8]	88.9%	100%	94.1%
$\geq 1.7\times$	6	1	2	7	81.3% [54.4, 96.0]	85.7%	75.0%	80.0%
$\geq 2.0\times$	6	1	2	7	81.3% [54.4, 96.0]	85.7%	75.0%	80.0%
$\geq 3.9\times$	6	0	2	8	87.5% [61.7, 98.4]	100%	75.0%	85.7%

Table 9: Walker Walk baseline seeds at 105K and 505K steps.

Seed	Eval@105K	Eval@505K	ve@65K	ve@105K
s1	915.7	970.4	0.563	0.656
s2	443.5	945.7	0.898	0.742
s3	869.0	947.7	—	0.664
s4	922.8	938.9	—	0.624

Table 10: Leave-one-environment-out cross-validation of NVE threshold $\theta = 1.5\times$. For each fold, the threshold is validated on an environment whose data was *not* used to select θ. The optimal threshold range on each training set is reported; $\theta = 1.5$ falls within the optimal range in all three folds.

Held-out	N_{train}	N_{test}	θ_{opt} range	Test Accuracy	Test AUROC
PickCube	18	2	$(1.45,\ 1.53]$	2/2 (1.000)	1.000
Cheetah	15	5	$(1.45,\ 1.53]$	5/5 (1.000)	1.000
Walker	7	13	$(1.00,\ 3.78)$	12/13 (0.923)	0.944
Mean	—	—	—	**0.974**	**0.981**

Table 11: dt-embedding gradient conditions on Walker Walk. Any gradient pathway updating e_{dt} is sufficient to induce CIVO; freezing or removing the embedding dramatically reduces severity.

Condition	Grad flow	Exp(s)	NVE peak	Eval	CIVO	
Trained	all	008d	1.53–4.62×	64–268	4/4	
Reward detach	obs+KL only	044	4.45×	44→538[†]	1/1	[*]Full-SGS
Full SGS	KL only	018	1.66×	448	1/1[*]	
Frozen	none	042, 015b, 047, 057	0.19–1.88×[‡]	204–612	2/4	
Removed	N/A	008f	0.97–1.41×	307–503	0/3	

seed 1 (NVE 1.66×); seed 2 collapsed. [†]Self-corrected at ∼175K (eval 44→538, 59% baseline; 224K run).
[‡]Exp_047 (NVE 1.71×) and Exp_057 (NVE 1.88×): both transient, self-correcting within 20K steps.

Table 12: Factorial design. Each row specifies one experimental configuration. **Emb**: dt-embedding mode. **Sub**: temporal subsampling. **H**: imagination horizon. **Imag** dt: temporal resolution used during imagination.

ID	Purpose	Emb	Sub	H	Imag dt
001	Reference baseline	—	—	15	1
006	Fixed $\mathrm{dt}{=}2$	trn	on	15	2
007	Fixed $\mathrm{dt}{=}4$	trn	on	15	4
008a	Unexposed emb ctrl	exp[a]	off	15	2
008c	Horizon control	exp[a]	off	8	2
008d	Collapse condition	trn	on	8	2
008e_v3	$\mathrm{dt}{=}1$ imag control	trn	on	15	1
008f	Causal ablation	—	on	8	2
015b	Frozen-emb ctrl	frz[b]	on	8	2
016e	Stop-grad (reward)	trn[†]	on	8	2
018	Full-SGS	trn[‡]	on	8	2
005	Learned gate	adaptive	off	15	learned

trn = trained ($d{=}32$, exposed to $\mathrm{dt}{>}1$ data via subsample ON); exp[a] = trained ($d{=}32$) but never exposed to $\mathrm{dt}{>}1$ data (subsample OFF); frz[b] = frozen random projection ($d{=}32$, `requires_grad=False`); — = removed ($d{=}0$).
[†] `reward_dt_emb_detach=True`; [‡] all non-KL gradients to `dt_emb` detached.

Table 13: Causal isolation: e_{dt} configuration $\times$ multi-rate data exposure. The combination of trained embedding and multi-rate exposure is the sole predictor of CIVO in tested configurations. Neither factor alone is sufficient.

	Multi-rate data (ON)	Single-rate data (OFF)
Trained e_{dt}	exp_008d: **CIVO** NVE 4.12×, collapse	exp_008a: healthy NVE 0.93×
Frozen e_{dt}	exp_015b: grad. starvation peak NVE 1.45× (transient)	—
Removed e_{dt}	exp_008f: no CIVO NVE $\leq$0.97×	—

Table 14: RSSM fidelity at $\mathrm{dt}{=}1$ vs. $\mathrm{dt}{=}2$. Trained embedding (008d) produces $2.78\times$ reward gap; removing it (008f) eliminates the gap but degrades absolute prediction.

	obs_MSE			**reward_MAE**		
Config	$\mathrm{dt}{=}1$	$\mathrm{dt}{=}2$	Gap	$\mathrm{dt}{=}1$	$\mathrm{dt}{=}2$	Gap
Baseline	13.49	27.50	2.04×	0.0102	0.0159	1.56×
008d (trained)	41.31	53.67	1.30×	0.0568	0.1579	2.78×
008f (no emb)	8.01	22.04	2.75×	0.7469	0.7526	1.01×

Table 15: Signed reward error (predicted $-$ actual) on posterior-conditioned RSSM predictions. Negative values indicate underprediction. The trained-embedding model underpredicts at $\mathrm{dt}{=}2$, ruling out observation-level reward overprediction as the CIVO mechanism.

Config	dt	**Mean SE**	**Std**	**%Over**	**Actual**	**Pred**
Baseline	1	$+0.001$	0.023	39.6%	1.893	1.894
008d (trained)	1	$+0.018$	0.072	61.8%	0.361	0.379
008d (trained)	2	$\mathbf{-0.599}$	0.722	8.0%	0.721	0.121
008f (no emb)	1	-0.710	0.704	14.0%	0.921	0.211
008f (no emb)	2	-1.645	1.485	5.0%	1.840	0.195

Table 16: Gradient contributions to the temporal embedding $e_{\Delta t}$ by loss component, measured via `torch.autograd.grad` with training configuration `gate_fixed_dt=2`. Gradient norms are ℓ_2 norms of $\nabla_\theta \mathcal{L}_i$ for each loss $\mathcal{L}_i$.

	@55K		**@112K**	
Loss component	Share	(%)	Share	(%)
Reward head	—	49.0	—	38.1
KL dynamics	—	34.7	—	40.4
Observation decoders	—	16.3	—	21.5

Table 17: Post-CIVO outcomes for all six CIVO-positive seeds across two environments. Peak/Final NVE measured in the 35K+ window.

Env	**Seed**	**Peak**	**Final**	**Eval**	**Outcome**
Walker	s1 (008d)	4.12×	4.12×	82.8	Full collapse
Walker	s2 (009b)	1.62×	0.38×	267.9	Transient CIVO
Walker	s3 (008d_s3)	1.53×	1.53×	268$^\diamond$	Late-onset collapse
Walker	s4 (008d_s4)	4.62×	4.62×	63.6	Full collapse
Cheetah	s1 (010b)	8.27×	8.27×	51.7	Full collapse
Cheetah	s0 (010b_s0)	108.52×	108.52×	0.83	Full collapse

$^\diamond$ Transitional; train returns still declining at 105K.

Table 18: Failure and degradation mode taxonomy. Each mode is characterized by its NVE trajectory, entropy behavior, and primary mechanism.

Mode	Representative	NVE	Entropy	Mechanism
Monotonic CIVO	008d, 008d_s4, 010b_s0	↑ sustained	↓	Reward bias → critic → actor
Late-onset CIVO	008d_s3	↑ delayed	↓	Same, delayed onset
Transient CIVO	009b	Peak → ↓	Stable	Self-correction via capacity
IRD collapse	008e_v3_s2	↑ late	↑	Entropy-driven, values stable
Grad. starvation	015b	Oscillates, decays	Stable	Frozen emb → zero actor grad
SGS-resistant	018_s2	↑	↓	Forward-pass conditioning
SGS-paradoxical	016e	↑ sustained	↓	Reward SGS → obs-dec reliance

Table 19: TCN results across environments and noise magnitudes. Prevention = peak NVE $< 1.5\times$ (35K+).

Env	σ	Seed	$\text{NVE}_{\max}$	Prevented?	Eval (%)
Walker	0.1	s0	1.02	✓*	55%
Walker	0.2	s0	0.91	✓	91%
Walker	0.2	s1	0.64	✓	45%
Walker	0.2	s2	2.11†	×	∼5%
Walker	0.2	s3	3.04¶	×	11%
Walker	0.5	s0	3.97	×	27%
Walker	0.5	s1	1.64	×	41%
Cheetah	0.2	s0	1.90	×	crashed
Cheetah	0.5	s0	10.67	×	26%
Cheetah	1.0	s0	≤0.30	✓§	15%
PickCube	0.2	s0	3.71	×	failing

*Borderline: NVE=$1.02\times$ at 95K, below $1.5\times$ threshold but above $1.0\times$.

†NVE $= 2.11\times$ at 95K, still rising; eval $\sim 5\%$ of baseline.

¶Late-onset CIVO: NVE declined to $0.67\times$ at 95K before phase-transition-like spike to $3.04\times$ at 105K; eval 349.6 at 95K → 71.8 at 105K → 104.5 at 215K.

§CIVO prevented (NVE$\leq$0.30); peak eval 116% baseline at 35K, declining to 15% by 215K via progressive policy erosion. See "impossible dose" discussion.

Table 20: DreamerV3 hyperparameters for all experiments. Modifications from defaults are marked with $\diamond$.

Parameter	Value
World Model	
RSSM discrete states	32 classes $\times$ 32 categories
RSSM stochastic dim	32
RSSM deterministic dim	512
MLP hidden size	512
MLP layers	2
Activation	SiLU
Normalization	LayerNorm
Training	
Batch size	16 sequences $\times$ 64 steps
Learning rate	3×10^{-4} (Adam, ϵ=10^{-8})
Sequence length	64
Training ratio	512
Free nats	1.0
KL balance (α)	0.8
Discount (γ)	0.997
λ-return	0.95
Policy	
Imagination horizon$^\diamond$	8 or 15 (see Table 3)
Actor entropy scale	3×10^{-4}
Actor gradient	Reinforce + dynamics backprop
Temporal Abstraction$^\diamond$	
Gate enabled	True (all non-baseline)
Gate fixed dt	1 or 2 (per config)
Subsample max dt	1 (OFF) or 2 (ON)
dt-embedding dim	32
Environment	
Action repeat	2 (Walker Walk, Cheetah Run)
Observation	Proprioceptive (no pixels)
Total environment steps	105K
Evaluation frequency	Every 10K steps
Evaluation episodes	10

Table 21: Retrospective CIVO detector validation (n=14). Detection type: "predictive" = alert precedes 20% eval decline; "reactive" = alert follows decline onset.

Exp	Config	Verdict	Trigger	Type	Peak Eval	r_{ve}
008d s0	trained dt-emb	TP	45K	predict.	167	0.69
010b	trained dt-emb (Cheetah)	TP	65K	reactive	322	3.18
008d s3	trained dt-emb	TP	45K	predict.	452	0.50
008d s4	trained dt-emb	TP	105K	reactive	383	3.03
010b s0	trained dt-emb (Cheetah)	TP	95K	reactive	248	6.34
008f s3	no dt-emb	TN	–	–	594	–
009b s2	trained dt-emb (self-corr.)	TN	–	–	279	–
baseline s1	vanilla	TN	–	–	970	–
009d s0	vanilla	TN	–	–	951	–
009d s1	vanilla	TN	–	–	920	–
019b	trained dt-emb (PickCube)	TN	–	–	26.6	–
009c s2	no dt-emb	FP	105K	reactive	359	0.75
009a s2	vanilla	FP	85K	predict.	967	0.73
009c s3	no dt-emb	TN	–	–	457	–
TPR = 5/5 (1.00), FPR = 2/9 (0.22)						

Table 22: Dual-critic 2×2 factorial results. Walker Walk unless noted.

Exp	Config	Params	NVE	Eval	Outcome
008d	Single narrow	1.2M	4.12×	82.8	Collapse
033	Wide single	2.4M	4.75×	407.9	CIVO → self-repair
035 s0	Narrow dual (min)	1.2M	2.75×	697.8	CIVO → recovery
039 s1	Narrow dual (min)	1.2M	2.19×	592.7	CIVO → recovery
032 s0	Wide dual (min)	2.4M	≤1.1×	739.7	CIVO prevented
032b s1	Wide dual (min)	2.4M	1.4×	551.8	CIVO prevented
038 s2	Wide dual (min)	2.4M	≤0.3×	120.9*	Non-CIVO collapse
Cross-environment					
034	Cheetah dual	2.4M	1.75×	97.5	Partial recovery (31%)
037	Cheetah wide single	2.4M	3.60×	146.1	CIVO → self-repair
036	PickCube dual	2.4M	2.3×	27.7	Transient CIVO

*CIVO prevented (NVE $\leq 0.3\times$) but non-CIVO eval collapse: eval peaked at 354 (155K), then declined to 120.9 (224K) with actor entropy increasing from -3.59 to -0.45.